

Using Machines to Advance Better Models of the Crypto Return Cross-Section*

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Abstract

Our analysis of the crypto market is centered around three questions: (i) What factors can explain the performance of different crypto portfolios? (ii) Which of these factors can be considered as priced risk factors? (iii) Can these factors help explain the observed average returns? We propose a cross-section approach that considers 33 portfolios and 13 factors based on multiple characteristics. By utilizing machine-learning techniques, we test and develop new models, while also addressing challenges such as the complex relationships between characteristics, the number of factors, the presence of similar characteristics, and the relevance of selecting factors with economic significance.

Keywords: Cryptos, machines, estimations, model selection, average crypto returns, cross-sectional pricing errors

JEL classification codes: G12, G13, G14, G24, G32.

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1. Introduction

Our contribution is to introduce a reliable factor model for evaluating expected returns in the crypto market. This version builds upon benchmarks for understanding the cross-section of crypto returns. Our model, called the crypto-eight-factor model, captures the essence of the cross-section of crypto returns. Not only does it provide a more accurate measure of expected crypto returns, but it also offers a conceptual framework that explains a wide range of potential findings in a cohesive manner. A challenge in asset pricing is to construct models that can explain the cross-section of expected crypto asset returns.

Many studies in financial economics focus on identifying factors that drive market fluctuations based on theories. However, we employ the commonly utilized portfolio approach to create a practical factor model, which is advantageous due to its simplicity and the availability of returns data. The proposed model's strong performance, along with its intuitive economic interpretation, makes it applicable to various practical uses such as assessing the performance of crypto-oriented funds, measuring abnormal returns in event studies, determining expected returns for portfolio selection, and estimating the cost of equity for capital budgeting, as well as possibly determining risk premiums for options on crypto futures. Cryptos are still opaquely understood.

Positioning our contribution. Our inquiry on the crypto market focuses on understanding the following aspects: (i) What are the contributing factors to the performance of different crypto portfolios? (ii) Can any of these factors be considered as key risk factors? (iii) How do these factors explain the average returns in the crypto market? To answer these questions, we analyze 33 portfolios (relying on 5,389 coins and tokens) and consider 13 factors based on various characteristics. To overcome challenges, such as the interrelationship between characteristics, a large number of factors, presence of similar characteristics, and relevance of selecting factors with economic significance, we utilize machine-learning techniques in our investigation and develop new

models. Our sample period is January 3, 2014, to January 26, 2024, with data on 525 weekly cross-sections of 33 crypto portfolio returns.

We develop a reliable indicator of crypto returns by combining a diverse group of potential predictors into a stochastic discount factor (SDF) representation. Our approach is able to predict out-of-sample performance of crypto returns in a high-dimensional setting by using a method that maximizes out-of-sample R^2 using a machine-learning algorithm subject to sparsity constraints. While previous research has mostly focused on SDFs of the crypto market with a few characteristics-based factors, our findings combine a large pool of potential factors to effectively approximate the candidate SDF. We then estimate SDF loadings and factor risk premiums. Furthermore, we propose test procedures for model comparisons and conduct alpha tests.

Our findings and value-added relative to the crypto literature. Based on a study conducted by Liu, Tsyvinski, and Wu (2022), the three-factor model for characterizing crypto returns includes the overall crypto market, size, and momentum. These three factors are found to have a significant impact on the expected weekly returns of cryptos over January 2014 to July 2020 (339 weeks). Through their analysis, they identify ten long-short strategies based on crypto characteristics. These featured strategies can be explained and understood through the lens of the three-factor model. However, our analysis shows that an eight-factor model outperforms the three-factor model in predicting average returns. Our empirical strategy is twofold. First, we employ a machine-learning approach that regularizes the sensitivity coefficients in the excess return regressions. Second, we employ a machine-learning approach that regularizes the SDF loadings. Both approaches build crypto models that are shown to improving on the three-factor model.

The analysis of crypto assets in Cong, Karolyi, Tang, and Zhao (2022) focuses on value and network adoption risk premiums in a factor-based framework. Furthermore, they present a categorization of major cryptocurrencies, allowing for an examination of token pricing both within

and across categories. Their findings reveal market segmentation in the crypto market, validating the economic functionality-based categorization, and providing insights in the pricing crypto assets. Their empirical analysis is documented over January 2014 to January 2021 (366 weeks).

The work of Dobrynskaya and Dubrovskiy (2020) analyzes different factors that could potentially influence the returns of cryptocurrencies and equities and determine which ones are responsible for driving returns and risk premiums. They find that only downside equity market risk, crypto size, and policy uncertainty factors consistently have significant premiums. Their analysis considers the sample period from January 2014 to November 2020 (356 weeks).¹

Deviating from these studies, our investigation utilizes a distinct methodology. We employ a three-stage process, starting with a minimizing criterion to determine the best set of beta coefficients taking into account pricing errors and regularization. These optimal coefficients are then tested on a separate out-of-sample dataset to calculate the out-of-sample R^2 . This process is automatized for cross-validation, where the coefficients are obtained from four folds and tested on the remaining fold, for a total of five cross-validation tests. The algorithm utilizes a range of penalty ratios to determine the most effective pair, which maximizes the out-of-sample R^2 . Each combination of factors is evaluated using this algorithm. In the second stage, we select the set of factors that result in the highest R^2 . Finally, in the third stage, we compare various crypto pricing models and estimate the SDF loadings, ultimately deducing support for an eight-factor model.

This paper highlights the importance of data characteristics in crypto asset pricing. The success of crypto investing strategies and availability of test assets for these models relies on understanding these characteristics. Additionally, digital currencies are becoming ingrained in the

¹Our research brings a unique perspective, and to better understand our contribution, it is important to consider how we compare to previous studies. We recognize that previous research has also explored factor-based asset pricing, but our findings highlight potential limitations in their approaches. For example, our dataset is more comprehensive and our consideration of multiple factors may offer a more complete view of the crypto market. Furthermore, our model has shown better performance when compared to previous ones. By providing this context, we can explain why studies may not have reached the same conclusion as ours, thereby emphasizing the significance of our findings.

investing realm and increasingly more financialized. However, the market environment post-COVID-19 have exposed a problem of missing data in crypto markets. As a result, the findings and techniques presented in this paper have implications beyond solely crypto pricing. The issue of new data presents an opportunity to develop more effective models that adequately capture the cross-section of returns in the dynamically expanding crypto market.²

Connection to recent methods in the asset pricing literature. Kozak, Nagel, and Santosh (2020) develop an estimation approach that minimizes the Hansen and Jagannathan (1997) distance, using L_1 and L_2 regularization penalties. Their goal is to obtain a set of SDF loadings that would increase out-of-sample R^2_{os} within a specific range of penalties. **Our method**, on the other hand, focuses on obtaining a sparse representation of the SDF by eliminating redundant traded factors in a sequential manner. Specifically, we discipline machine-learning algorithms in conjunction with factor selection that combine minimizing the mean-squared error of many (potentially large) factor models, cross-validation, and the R^2 criterion.

To manage the abundance of factors, newly added factors are viewed with skepticism. In this regard, Feng, Giglio, and Xiu (2020) present a model-selection technique that evaluates the impact of an introduced factor on asset pricing, in comparison to a high-dimensional set of known factors. Their work takes into consideration potential errors in model selection that can lead to confounding results by omitting relevant variables. We supervise machine-learning alongside traditional estimation methods to highlight both statistical and economic relevance of factor models.

Research on empirical asset pricing has revealed a growing number of factors being used to explain returns, and some of these generated factors may not accurately capture the underlying risk sources. However, a solution has yet to be proposed that can address all of these issues, such

²For a partial list of the crypto market research, see, among others, Griffin and Sham (2020), Liu and Tsyvinski (2020), Bianchi and Babiak (2021), Zhang, Li, Xiong, and Wang (2021), John, O’Hara, and Saleh (2022), Fan, Feng, Lu, and Tong (2023), Huang, Lin, Lu, and Sun (2023), and Cai and Zhao (2024).

as being able to handle tradable factors, the factor spectrum, misspecification, the weak inference problem, and providing a more valid measure of risk compared to established models. A Bayesian method is proposed by Bryzgalova, Huang, and Julliard (2023) to understand these challenges.

Giglio and Xiu (2021) consider a three-pass approach for estimating the risk premium associated with an observable asset factor. They use principal components of test asset returns and additional cross-sectional and time-series regressions to identify the risk premium of the observable factor. Their methodology also addresses potential measurement error in the observable factor and can identify when a factor is spurious or otherwise irrelevant. Acknowledging the relevance of many such concerns, our estimation approach accommodates the uncertainty associated with estimating the means and covariances of the traded factors.

2. Crypto characteristics: January 3, 2014, to January 26, 2024

Since crypto assets are traded across numerous global exchanges, collecting this information manually is impractical. We utilize coinmarketcap.com, a reputable source in the crypto market, which compiles data from over 200 major exchanges. The dataset used in our study consists of all actively traded cryptos listed on coinmarketcap.com as of January 26, 2024. See <https://coinmarketcap.com/methodology> for CoinMarketCap’s selection methodology. Our dataset includes crypto coins and digital tokens, as both are considered tradable assets.

We begin in 2014 because it marks the start of significant crypto trading activity. Our data sample is composed of weekly aggregates (525 weeks), starting from Friday and ending on the following Friday. This dataset includes the entire duration of the COVID-19 pandemic, allowing for analysis of potential changes in crypto trading patterns before, during, and after the pandemic.

To be selected for our study, a crypto must have had complete data on its price, trading volume, and market capitalization. No further criteria, such as a minimum market capitalization, are

imposed, as this information is already incorporated by coinmarketcap.com. We also make use of winsorization to adjust for potential data-recording mistakes, setting the upper limit at the 99.5th percentile for each crypto price, as well as its capitalization and volume.

Table 1 displays the changing characteristics of the crypto market. The initial sample consists of 55 coins and three tokens in the first weekly cycle, while the final cycle includes 551 coins and 4,838 tokens. There is a noticeable upward trend in the average market capitalizations of cryptos, with coins increasing from an average of \$138 million to \$4.5 billion, and tokens rising from \$3.7 million to \$43.6 billion. However, the average trading volume reaches its peak in 2021, suggesting a possible shift in crypto behavior after the pandemic.

Mean circulation rates for cryptos are also shown in Table 1. The circulation rate is a measure of the amount of crypto available for trading. A lower circulation rate indicates that a portion of the crypto is being held back, often by the creators. The data reveals lower circulation rate values between 2018 and 2020, with tokens consistently below those for coins. This difference may be attributed to the nature of these assets: Coins function more like traditional currencies, while tokens resemble non-currency assets in terms of investment characteristics.

Table 2 provides information on crypto returns by examining the excess return for each crypto. This calculation is $r_{t+\Delta}^{e,i} \equiv (1 + r_{t+\Delta}^i) - R_{t+\Delta}^{\text{rf}}$, where $r_{t+\Delta}^i = \frac{S_{t+\Delta}^i}{S_t^i} - 1$ and S_t^i is the price of crypto asset i at time t . Additionally, $R_{t+\Delta}^{\text{rf}}$ is the gross risk-free return over Friday-to-Friday (known at time t) and $\Delta = \frac{1}{52}$.

According to Table 2, the equal-weight excess return, which evenly averages crypto returns across successive Friday closing values, is 3.5%. On the other hand, the value-weight excess return, which takes into account each crypto's market capitalization, is lower, at 1.9% per week. The weekly volatility of the value-weight excess return is considerable, at 19.1%, implying the high-risk nature of crypto investments.

Distinct patterns emerge when analyzing coins and tokens, each with their own unique risk profiles. Value-weight tokens have an average of 3.1% excess return per week, while coins have an average of 1.0% excess return. Additionally, tokens tend to have higher return volatility with a weekly standard deviation of 27%, while coins have a standard deviation of 11%. The market capitalization distribution is skewed toward the four prominent cryptos (Bitcoin, Ethereum, Ripple, and Dogecoin), which make up an average of 46% of the total crypto market. Our objective is to effectively price a variety of crypto assets by eliminating redundant factors and comparing multiple promising models.

3. Machines and dissecting the cross-section of crypto returns

We provide a framework for analyzing asset pricing models. Through this analysis, we show how to estimate factor risk premiums and SDF loadings, evaluate hypothesis tests, and create confidence intervals for model coefficients and alphas. One aspect of our approach is that it accounts for cases where there are multiple nonnested misspecified SDFs. In these situations, our method can identify the dominant factor model that subsumes the effect of other models.

3.1. *Estimation of crypto models and construction of factors and test assets*

Asset pricing theories focus on explaining the expected returns of portfolios based on various risk factors. In this framework, the vector of excess returns for each test portfolio, $\mathbf{r}_{t+\Delta}^e$, comprises the excess return of test portfolio i from t to $t + \Delta$. This depiction allows for an understanding of the underlying drivers of portfolio returns and for the evaluation of the effectiveness of different risk factors in explaining return cross-sections.

Consistent with established theory, we formulate the asset pricing relations, as follows:

$$E [\mathbb{m}_{t+\Delta} \mathbf{r}_{t+\Delta}^e] = \mathbf{0}, \quad \text{with} \quad \mathbb{m}_{t+\Delta} = 1 - \mathbf{b}'(\mathbf{f}_{t+\Delta} - \boldsymbol{\mu}). \quad (1)$$

In (1), $\mathbb{m}_{t+\Delta}$ represents the SDF, $\mathbf{f}_{t+\Delta}$ a set of risk factors, and $\boldsymbol{\mu}$ the factor means. The vector of parameters, $\mathbf{b}$, also appears in an SDF formulation of the pricing equation (1). These parameters — the SDF loadings — are estimated via a generalized method of moments (GMM), as follows:

$$E \begin{bmatrix} (1 - \mathbf{b}'(\mathbf{f}_{t+\Delta} - \boldsymbol{\mu})) \otimes \mathbf{r}_{t+\Delta}^e \\ \mathbf{f}_{t+\Delta} - \boldsymbol{\mu} \\ \text{vec}((\mathbf{f}_{t+\Delta} - \boldsymbol{\mu})(\mathbf{f}_{t+\Delta} - \boldsymbol{\mu})') - \text{vec}(\Omega_{\mathbf{f}}) \end{bmatrix} = \mathbf{0}, \quad (2)$$

where $\Omega_{\mathbf{f}}$ is the variance-covariance matrix of $\mathbf{f}_{t+\Delta}$. We follow the methodology of studies such as Jagannathan and Wang (2002) and Menkhoff, Sarno, Schmeling, and Schrimpf (2012). This method allows us to incorporate the uncertainty associated with estimating the means and covariances of $\mathbf{f}_{t+\Delta}$.

The specification of the SDF in (1) implies that it can be represented using a beta model. In this approach, the expected excess returns of each asset are determined by a factor risk premium vector $\boldsymbol{\lambda}$, and the asset-specific risk loadings represented by vector $\boldsymbol{\beta}_i$. In other words,

$$E[r^{e,i}] = \boldsymbol{\lambda}'\boldsymbol{\beta}_i, \quad \text{where} \quad \boldsymbol{\lambda} = \Omega_{\mathbf{f}}\mathbf{b}. \quad (3)$$

We examine results based on the one-step GMM (which uses the identity matrix as a weighting matrix) and those based on the two-step GMM (which uses the optimal weighting matrix).

To implement the GMM estimation in (2), we analyze 13 factors, resulting in a pool of many possible SDF models. Each factor construction is based on a long and short portfolio combination based on a characteristic. These characteristics include size, reversal, momentum, attention, liquidity, lottery, circulation, skewness, coskewness, downside risk, upside potential, and golden cross. See the method of the factor construction in Table 3 and the corresponding summary statistics in Table 4. The long and short legs of these strategies are chosen to generate a positive average

return for each factor.

Our findings suggest that only a sparse set of these factors can be considered reliable indicators of cross-section of crypto returns. Through our analysis, we show that there are definitive models that stand out among all the potential combinations of factor investments. In fact, a seven- to nine-factor model consisting of the most promising factors outperforms the three-factor benchmark due to Liu, Tsyvinski, and Wu (2022). Specifically, their model favors market, size, and momentum.

We create a dataset of crypto portfolios for the purpose of testing crypto models and generating model representations. This dataset consists of 33 portfolios, each representing a different segment of the crypto market based on specific characteristics. This provides a diverse and potentially representative sample of the market for our analysis.³ These portfolios are based on Decile 1, Decile 5, and Decile 10 of various characteristics, encompassing size, reversal, momentum, attention, liquidity, lottery, skewness, coskewness, downside risk, upside potential, and golden cross. We allow for an analysis of various factors and their impact on crypto portfolios. By including a range of portfolios, we aim to capture broad behaviors in order to draw meaningful comparisons and inform investment strategies. The details of each test portfolio are in Table 5.

Table 4 (bottom panel) displays the correlations between the crypto factors. The results show that there is generally low correlation among the factors, with 17 out of 78 absolute correlations above 0.25 (indicated by bold font). This implies that the factors reflect distinct characteristics. However, there are some notable correlations. The market factor ($f_{t+\Delta}^{\text{market}}$) is strongly negatively correlated with the size factor ($f_{t+\Delta}^{\text{size}}$), with a coefficient of -0.75 , suggesting that crypto $f_{t+\Delta}^{\text{market}}$ is closely linked to the size of the crypto assets being traded. Additionally, there is a strong positive correlation of 0.75 between the attention factor ($f_{t+\Delta}^{\text{attention}}$) and the liquidity factor ($f_{t+\Delta}^{\text{liquidity}}$). This

³The problem of inadequate test assets in empirical asset pricing has been examined, including recently by Bryzgalova, Pelger, and Zhu (2024) and Giglio, Xiu, and Zhang (2024). It is crucial to select suitable test assets, as they need to represent the underlying SDF. In line with this notion, we incorporate a wide variety of test assets, as the appropriate selection of these assets is essential not only for assessing models but also for constructing factors.

feature indicates that attention — or interest in a particular crypto — is associated with higher trading volume, alluding the possible impact of investor sentiment.

3.2. *Nonsparse SDFs and disciplining model constructions*

The concept of sparsity is frequently utilized in asset pricing, although it may not always be explicitly acknowledged. In our scenario, we are required to estimate and assess 8,100 ($= \sum_{r=3}^{13} \frac{13!}{(13-r)!r!}$) factor models. From this perspective, sparsity refers to incorporating a limited number of factors in an empirical model of SDF, its corresponding beta model, and underlying time-series return representations.⁴

We adopt the elastic-net framework with cross-validation to initially guide the selection of the best combinations of certain number of factors, for example, selecting eight from a list of 13 in a model. The following problem is used to achieve sparsity in the asset pricing models:

$$\min_{\boldsymbol{\beta}} \frac{1}{2T} \|\mathbf{r}^{\text{e,actual}} - \mathbf{f}\boldsymbol{\beta}\|^2 + \boldsymbol{\psi} \times w \times \|\boldsymbol{\beta}\|_{21} + \frac{\boldsymbol{\psi} \times (1 - w)}{2} \times \|\boldsymbol{\beta}\|^2, \quad (4)$$

where $\|\cdot\|$ indicates Frobenius norm, $\|\boldsymbol{\beta}\|_{21} \equiv \sum_{k=1}^{13} \sqrt{\sum_{i=1}^{33} \beta_{ki}^2}$, and T is the number of time-series observations (i.e., 525 weeks). Additionally, $\boldsymbol{\psi}$ and w align with penalty ratios. Notably, the algorithm shrinks to LASSO (ridge) regressions when $\boldsymbol{\psi} = w = 1$ ($\boldsymbol{\psi} = 1$ and $w = 0$).

For our purposes, we employ a fivefold cross-validation approach to estimate the parameters $\boldsymbol{\beta}$ (along with $\boldsymbol{\psi}$ and w) subject to factor selection. Specifically, for a given set of $\mathbf{f}$, $\boldsymbol{\psi}$, and w , the algorithm first estimates $\boldsymbol{\beta}$ by solving (4). This cross-validation technique divides the dataset into five equal-sized folds, training the model on four folds, and validating it on the remaining fold. The accuracy of the fit is assessed using the mean-squared-error on the validation fold. This

⁴Sparse models are often preferred because they offer a more amenable representation of expected returns and tend to outperform more complex models in out-of-sample exercises. Additionally, sparse models are easier to interpret and align with economic theories compared to factor models which use principal component aggregation. It is beneficial to prioritize sparsity in modeling since dense models may overfit and can be challenging to justify.

process is repeated five times, with each fold serving as the validation set in turn, and the average mean-squared-error score is computed across all iterations.

Subsequently, the algorithm repeats this process over a grid of specified values for ψ and w to identify the optimal combination that yields the highest fitting accuracy. Once the optimal values of ψ and w are determined, the algorithm resolves the minimization problem defined by (4) using the entire dataset to obtain the optimal β^{optimal} and the corresponding accuracy score for this given set of factors $\mathbf{f}$,

$$R^2_{\text{elastic-net}}[\mathbf{f}] \equiv 1 - \frac{\|\mathbf{r}^{\text{e,actual}} - \mathbf{f}\beta\|^2}{\|\mathbf{r}^{\text{e,actual}} - \bar{\mathbf{r}}^{\text{e,actual}}\|^2}. \quad (5)$$

Next, to identify the optimal combination of factors, $\mathbf{f}$, we apply the entire algorithm to every possible combination of factors (e.g., selecting eight factors from a set of 13). We then select the combination, $\mathbf{f}^{\text{best}}$, that yields the highest $R^2_{\text{elastic-net}}[\mathbf{f}^{\text{best}}]$.

We have chosen *scikit-learn*, a Python machine-learning library, to implement elastic net. This particular library provides the option to generate a grid of ψ -values automatically. As such, we need to define a list of only w -values for the calculations, such as [0.01, 0.05, 0.1, 0.2, 0.3, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 0.99]. Various studies, including those by Freyberger, Neuhierl, and Weber (2020), Gu, Kelly, and Xiu (2020), and Kelly and Xiu (2023), have utilized different approaches to incorporate elastic net into their research on asset pricing.⁵

The resulting asset pricing models, featured in Table 6, are built upon selected nonzero β and SDF loadings $\mathbf{b}$, include ten versions of the encompassing model estimated. These models, obtained using the elastic net (LASSO or ridge) lead to predetermined combinations of factors.

Our regularization procedures indicate that the circulation factor never passes model selection. Additionally, the factor model most frequently comprises market (11 times), size (11 times),

⁵As opposed to our elastic net approach with cross-validation, Feng, Giglio, and Xiu (2020) propose a double-selection LASSO method. Their aim is to address concerns over traditional factor selection techniques that potentially lead to model selection errors and introduce bias through omitted variables.

lottery (11 times), momentum (nine times), reversal (nine times), downside risk (eight times), and upside potential (eight times) factors. The essence of this selection procedure is to generate sparsity for the returns model, while acknowledging the isomorphism between the beta model and the corresponding SDF representation.

3.3. *Cross-sectional variation in average excess returns and links to risk factors*

Motivated by our evidence in Table 6, we next discipline model constructions by computing performance statistics in Table 7 and conducting GMM estimations of SDF loadings, $\mathbf{b}$, using (2). Then we assess the effectiveness of SDF representations in explaining the variation of returns in various crypto portfolios. By using this approach, we conduct a comparison between the various three-factor crypto models and more complex models with up to 13 factors. Overall, our methodology addresses the suitability of factor models in explaining the return cross-section of crypto portfolios. We investigate both in-sample and out-of-sample model performance.

3.3.1. *A snapshot of performance across the 14 models*

Table 7 summarizes the performance of different models, ranging from three factors to 13 factors. Our focus is on how adding more crypto factors affects the benchmark three-factor model of Liu, Tsyvinski, and Wu (2022). Our results suggest a bowl-shaped pattern in the average absolute alphas ($A^{|\alpha|} \equiv \frac{1}{N} \sum_{i=1}^N |\alpha_i|$), with portfolio α 's obtained from time-series regressions. The highest $A^{|\alpha|}$ is obtained by the three-factor model with market, size, and momentum, which gradually decreases as more factors are added up to the eight-factor model. After that, we see an increase in $A^{|\alpha|}$ for the dense 13-factor model. The turning point of $A^{|\alpha|}$ is 1.11%, reached by the factor model supported by both the elastic net and ridge procedures.

The overall results show consistent evidence that the eight-factor model derived from elastic net and ridge procedures — the crypto-eight — provides a more accurate representation of average returns for different types of portfolios compared to alternative models. This can be seen through

various measures, such as the Hansen and Jagannathan (1997) distance measure, denoted HJ_{distance}^+ , which shows a noticeable improvement with the eight-factor model. The “ $100 \times HJ_{\text{distance}}^+$ distance” column displays the measure, which indicates the normalized maximum pricing errors. When considering the three-factor model, the $100 \times HJ_{\text{distance}}^+$ is 9.92. On the other hand, the eight-factor model has a $100 \times HJ_{\text{distance}}^+$ of 8.88. This implies that the eight-factor model provides more precise pricing in comparison to the three-factor model driven by market, size, and momentum.

The $\bar{R}^2$ values for the 33 excess return portfolios are calculated for each factor model specification, and the average, denoted $A\bar{R}^2$, is determined. Our results indicate that the eight-factor model achieved a higher $A\bar{R}^2$ of 35.4% compared to the three-factor — based on market, size, and momentum — model’s $A\bar{R}^2$ of 25.5%. This implies that the eight-factor model is more successful in explaining the returns of various crypto portfolios. These findings highlight the effectiveness of the sparsity approach in identifying a robust model for analyzing and predicting returns across diverse portfolio types.

3.3.2. GMM estimation of models and economic and statistical relevance of crypto factors

Table 8 displays the estimated SDFs by first utilizing 33 test portfolios. Return data is Friday-to-Friday, and we obtain the $\mathbf{b}$ coefficients through two-step GMM procedure. All models are evaluated against the cross-section of crypto returns.

Allied to an estimated coefficient are superscripts “***” (Newey and West (1987) HAC p -value less than 0.05) and “*” (p -value between 0.05 and 0.1), which indicate the statistical significance of these factors in explaining the returns of the portfolios.

The analysis of results in Table 8 (Panel A) reveal that the SDF loading parameters associated with the market, size, lottery, and downside risk factors are positive and statistically significant, indicating an explanatory power for these factors in the cross-section of crypto returns. The market, size, lottery, and downside factors support positive risk premiums, suggesting that portfolios

with a higher covariance with these factors receive higher compensation.

Factors related to reversal, upside, and golden cross do not have a significant effect on the expected returns of the portfolios in the eight-factor model. Additionally, the parameters associated with the risk loading for these factors are also insignificant, possibly implying that these factors do not explain a significant amount of the variation in expected returns.

The findings of our study reveal that the three-factor model is weaker compared to alternative multifactor models in predicting crypto returns. In particular, while the SDF loading for market and size, and the associated factor risk premiums (λ), is significant, the momentum factor has lost significance in our sample up to January 2024. Overall, our investigations show the importance of market and size risk premiums, which suggests that market and size are significant risk factors that impact asset pricing.

Thus, our results show that the three-factor model, comprising market, size, and momentum factors, cannot fully explain the cross-section of crypto returns. To explore the potential contribution of other factors, we return to our two-step GMM estimation with an optimal weighting matrix. By testing exclusion restrictions on the SDF with many factors versus various three-factor models in Table 9, we address the issue of redundant factors. Our findings suggest that the exclusion restrictions of (i) $b_{\text{lottery}} = b_{\text{skewness}} = b_{\text{downside}} = b_{\text{upside}} = b_{\text{golden cross}} = 0$, (ii) $b_{\text{reversal}} = b_{\text{lottery}} = b_{\text{skewness}} = b_{\text{downside}} = b_{\text{upside}} = 0$, (iii) $b_{\text{reversal}} = b_{\text{lottery}} = b_{\text{skewness}} = b_{\text{downside}} = b_{\text{upside}} = b_{\text{golden cross}} = 0$, and (iv) $b_{\text{reversal}} = b_{\text{lottery}} = b_{\text{downside}} = b_{\text{upside}} = 0$ are statistically rejected. This indicates that additional factors, such as lottery, skewness, downside, upside, and golden cross, have economically significant loadings on the SDF and provide additional pricing flexibility in explaining the cross-section of crypto returns. The χ^2 statistics for these exclusion restrictions are all 0.000, reinforcing the importance of these additional factors in explaining crypto returns on top of these three-factor models.

We draw statistical inferences regarding models' HJ_{distance}^+ in Table 8 (Panel C). We block-

bootstrap factors and assets jointly with replacement and compute HJ distance in each bootstrap draw, for each model. We judge $HJ_{\text{distance}}^{+, \text{B Model}} < HJ_{\text{distance}}^{+, \text{A model}}$ by testing against hypothesis $H_0 : 100 \times \left(\frac{HJ_{\text{distance}}^{+, \text{A model}}}{HJ_{\text{distance}}^{+, \text{B Model}}} - 1 \right) \geq 0$ versus $H_A : 100 \times \left(\frac{HJ_{\text{distance}}^{+, \text{A model}}}{HJ_{\text{distance}}^{+, \text{B Model}}} - 1 \right) < 0$. We bootstrap the sample 10,000 times and compute the empirical p -value as the proportion that $100 \times \left(\frac{HJ_{\text{distance}}^{+, \text{A model}}}{HJ_{\text{distance}}^{+, \text{B Model}}} - 1 \right) < 0$. We also compute the White's heteroskedasticity-consistent one-sided p -value by running the series of 10,000 differences on a constant. Comparing with the empirical p -value, the White's p -value takes into account the sizes of the differences. This investigation shows that the three-factor model does not fare better in our comparisons than competing models.

Our conclusion of a seven- to nine-factor SDF is justified by various crypto data characteristics, including extracted principal components. The test assets exhibit a sizable correlation between the crypto market factor and the first principal component, explaining 30.3% of the total variation. The second principal component, which represents 9.3% of the variation, is heavily influenced by factors such as size, momentum, reversal, lottery, and coskewness. This analysis suggests that seven to nine factors are needed to capture a significant portion (67.5%) of the variability observed in the 33 baseline portfolios, despite the potential for 33 principal components.

3.3.3. Average returns of crypto portfolios implied by models

Figure 1 illustrates expected return pricing error by showing the model's pricing errors. The errors are represented by the deviations from the 45-degree line, indicating discrepancies in the predicted returns compared to the actual returns for some portfolios. Additionally, we calculate the fitted average returns using equation (3) and use the uncentered R^2 's and mean absolute errors (MAE_{OLS}) measures as indicators of the model's goodness-of-fit. The MAE_{OLS} is calculated as $(1/33) \sum_{i=1}^{33} |\text{Realized}_i - \text{Fitted}_i|$, in weekly percentage unit.

Table 10 reveals that the crypto portfolios with the smallest market capitalizations, as well as the smallest and middle portfolios based on reversal, liquidity, and lottery are the most challenging

to accurately price using the models. Despite these difficulties, these portfolios hold significant interest from a model perspective.

Using a GLS cross-sectional uncentered R^2 of 78.5% and an OLS uncentered R^2 of 80.1%, it can be concluded that the eight-factor model effectively captures a significant portion of the cross-sectional variation in cryptos. However, the χ^2_{NW} tests (with $N - K$ degrees of freedom (Cochrane (2005, 12.2, page 237))), which show p -values of 0.00 with Newey and West’s method, indicates that the asset pricing model is rejected for the featured 33 crypto portfolios.

3.3.4. *Removing insignificant SDF loadings in crypto models worsens performance*

The economic contribution of each crypto factor in explaining the cross-section of returns can be quantified by considering the issue of potentially redundant factors from two perspectives. First, in light of our findings from Table 8, we examine results from a restricted version of the eight-factor model that jointly imposes zero SDF loadings on (i) size, (ii) lottery, and (iii) downside factors (the ones that are significant). Our analysis shows that this restricted five-factor model is rejected. This evidence relies on the p -values of 0.000 for the $\chi^2(3)$ statistic, which indicates that omitting these factors has a negative impact on the model’s performance. Furthermore, the R^2_{GLS} drops and the Hansen-Jagannathan distance increases, indicating that the eight-factor model provides a better characterization of crypto returns compared to its nested counterparts. Therefore, the eight-factor generalization is an appealing approach for explaining the return cross-section.

Second, Kozak, Nagel, and Santosh (2020) show that incorporating a significant number of factors based on equity market characteristics is essential for achieving consistent performance in the SDF, even if the SDF loadings are not statistically significant. Our approach ensures that only economically relevant factors are included in the crypto SDF. The results in Table 11, notably based on $\text{HJ}^+_{\text{distance}}$, MAE_{OLS} , and $A^{|\alpha|}$, support our contention that crypto models with a restriction on the number of factors (eight or nine) perform worse compared to the original model with a

higher number of factors. This is done by excluding (i) reversal, (ii) skewness, (iii) upside, and (iv) golden cross in the eight-factor model (see Panel A of Table 11), or excluding (i) upside in the nine-factor model (see Panel B of Table 11).

In sum, this analysis highlights the value-added of incorporating a sufficient number of factors in the SDF, even if their SDF loadings may not be statistically significant. This part of our investigation suggests that the identified factors are not spurious, but likely reflect true risk factors driving crypto returns.

3.3.5. Downside risk factor and lottery factor are economically relevant

In their study, Lettau, Maggiori, and Weber (2014) demonstrate that incorporating a downside risk factor in addition to the market factor leads to improved pricing across traditional asset classes. Our seven, eight, and nine factor specifications in Table 8 consistently show strong significance in both the estimated factor risk premiums and risk loadings of the SDF, indicating the practical and statistical relevance of including a downside factor in crypto model.

The level of lottery preference in cryptos exerts an impact on model performance and behavior. Cryptos with a higher lottery preference tend to be smaller in size and have less liquidity. They also exhibit a stronger tendency for anomalies, such as crypto volatility. Compared to cryptos with a lower lottery preference, those with a high preference show more pronounced returns. This is substantiated by the significant and positive coefficients of both b_{lottery} and λ_{lottery} in Table 8, indicating the economic relevance of the lottery factor. These findings suggest that including a lottery factor in crypto pricing models can improve their explanatory power for these anomalies.

In summary, our study highlights the significance of downside risk in understanding crypto market behavior and underscores its relevance in pricing models. Our findings also suggest that investors are drawn to lottery-style investments, prompting them to invest in highly volatile cryptos, ultimately favoring their potential returns.

3.3.6. Alpha tests from the time-series regressions and spanning tests

We evaluate the performance of the models in capturing crypto returns by conducting regression analyses on the baseline test assets, $i = 1, \dots, 33$. The regressions are of the form

$$r_{t+\Delta}^{e,i} = \alpha^i + \sum_{k=1}^K \beta_{ki} \times f_{t+\Delta}^k + \varepsilon_{t+\Delta}^i \quad \text{for } t = 1, \dots, T. \quad (6)$$

We investigate individual significance on each α^i . Additionally, we test the adequacy of the models by assessing the hypothesis that $\alpha = [\alpha^1, \dots, \alpha^{33}]' = \mathbf{0}$. This hypothesis of zero pricing errors is examined using a GMM approach, where the statistic $\hat{\alpha}' \text{var}(\hat{\alpha})^{-1} \hat{\alpha}$ follows an asymptotic chi-square distribution with 33 degrees of freedom (Cochrane (2005, pages 233 and 234)).

The results, reported in Table 12, show that the eight-factor model can describe the returns variation in the 33 test portfolios. In particular, 20 out of the 33 α^i estimates have p -values that exceed 0.1 and the largest (absolute) α has a value of 6.7% weekly, while the majority of α 's are small, implying that the departures from the model are economically small. However, the p -value of 0.00 indicates that the null hypothesis of $\alpha = \mathbf{0}$ is rejected.

We draw the following additional conclusions. First, the β_{market} coefficients for crypto portfolios are consistently positive and significant, indicating that the returns of these crypto portfolios are influenced by the general crypto market movements. Further analysis reveals that the crypto portfolios exhibit a strong size effect, with the β_{size} coefficients consistently positive and significant, establishing the link between crypto portfolio returns and their size. Third, while the reversal factor does not significantly influence the returns of portfolios sorted by size, it does explain the returns of portfolios categorized by momentum, skewness, and upside.

This suggests that the size, reversal, and lottery factors are distinct and explain returns for different portfolios in the time-series. This further emphasizes the usefulness of considering a variety of distinct factors in constructing and analyzing crypto portfolios.

From a statistical perspective, the performance of the three-factor model is relatively unsatisfactory. Although not tabulated for each crypto portfolio, the $\bar{R}^2$ values are lower (see the average in Table 7). This is likely due to the model being incorrectly specified, as a considerable number of α^i are observed to be significantly different from zero, indicating a lack of adequacy. Overall, eight factors better capture the fluctuations in crypto returns.

We conduct factor spanning tests by performing time-series regressions of each factor on the baseline eight-factor model, following Fama and French (2015). The data in Table 13 imparts key takeaways. For one, the intercepts in the regressions involving market, size, momentum, coskewness, and downside, show p -values of 0.00. Thus, the intercepts — the returns explained by eight factors other than itself, remain statistically significant for five out of 13 factors, ranging from 0.32% to 4.61% per week. For another, the intercepts for the reversal and lottery factors manifest high p -values. However, our evidence on SDF loadings and HJ_{distance}^+ tests reveal an economic role for both reversal and lottery. For example, the lottery factor is statistically relevant in the presence of market and size factors in Table 8.

All in all, our analysis reveals that the eight-factor model consistently outperforms other models in explaining the crypto cross-section.⁶

3.3.7. *Our results are favorable relative to methods that distill factors through aggregation*

How does our approach with observable factors compare to an approach that distills the principal components from the factors? In analogy to extant approaches, we decompose the SDF specification into two parts, as follows:

⁶We emphasize two comparisons. First, the models used in this study have relatively lower explanatory power (the average $\bar{R}^2$ ($A\bar{R}^2$)) compared to equity studies such as those done by Fama and French (2015). Additionally, the cross-sectional χ^2 tests generate consistently low p -values of 0.000, indicating difficulties in accurately fitting the volatile returns of cryptos. This implies that certain crypto portfolio returns, particularly those related to size, reversal, and momentum, are not adequately captured by the combination of models used in our analysis.

$$\mathbb{m}_{t+\Delta} = \begin{cases} 1 - \mathbf{b}_{\text{market}}^* (f_{t+\Delta}^{\text{market}} - \mu_{\text{market}}) - \underbrace{\mathbb{m}_{t+\Delta}^\perp}_{\sum_{j=1}^{12} \mathbf{b}_j^* \times f_{t+\Delta}^{\text{pc}j}}, \\ 1 - \mathbf{b}_{\text{market}} (f_{t+\Delta}^{\text{market}} - \mu_{\text{market}}) - \mathbf{b}_{\text{mve}} (\mathbb{m}_{t+\Delta}^{\text{mve}} - \mu_{\text{mve}}). \end{cases} \quad (7)$$

Our findings support the effectiveness of our method versus the SDF that can be constructed using the market and 12 orthogonalized factor components ($f_{t+\Delta}^{\text{pc}j}$ for $j = 1, \dots, 12$). Additionally, here we do not identify sparsity, where a small number of portfolios and principal components are sufficient to represent the SDF, which aligns with characteristics of these portfolios. The approach to construct $\mathbb{m}_{t+\Delta}^{\text{mve}}$ follows the ridge regularization procedure of Kozak, Nagel, and Santosh (2020), who use principal components as factors to explain the returns of asset portfolios in the space of equity anomalies, net of market exposure.

In essence, these developments result in SDFs that considers both the market returns and nonmarket factors, whereby the principal components capture the majority of the variability in the crypto returns. On the other hand, $\mathbb{m}_{t+\Delta}^{\text{mve}}$ incorporates the weighting of risk factors rank-ordered by the estimated price of risk. To examine the dependability of our conclusions, we then estimate both SDF specifications in equation (7) by GMM.

Table 14 allows for two conclusions. First, like other models, the p -values for the Hansen's J -statistics are 0.00 and the overidentifying restrictions of the models are rejected. Second, the $100 \times \text{HJ}_{\text{distance}}^+$ obtained using the two-step (iterated) GMM are 14.90 (11.73), which, as noted from Table 7, are higher than those from the three-factor model and the crypto-eight model. Thus, the PC-based approaches manifest poor cross-sectional properties even though the in-sample fitting based on the R_{OLS}^2 and MAE_{OLS} fare better, implying the possibility of overfitting.

In sum, our analysis survives robustness check with respect to principal components, to the use of iterated GMM, and to the weighting of risk-factors with the highest price of risk. Our work

aligns with previous research that utilizes a broad sample of asset returns to analyze the risk-return relationship through factor models. However, our particular emphasis is on investigating specific aspects of these trade-offs in the crypto market.

3.3.8. Cross-sectional out-of-sample comparisons

Kozak, Nagel, and Santosh (2020) derive a Bayesian estimator of SDF loadings that entails minimizing the Hansen-Jagannathan distance subject to L_1 and L_2 penalties on SDF loadings. Our methodology does not depend on the Hansen-Jagannathan distance as a target, but rather focuses on minimizing pricing deviations subject to the regularization of sensitivities in the return time-series regressions via elastic net, LASSO, or ridge, and cross-validation. Overall, the two approaches share a goal of estimating the SDF but differ in their methods and underlying regularization.

Our analysis in this section uses both cross-sectional and time-series methods to evaluate the effectiveness of developed crypto asset pricing models through an out-of-sample approach. Our focus is on addressing two questions: (1) Is the three-factor model less reliable in out-of-sample exercises compared to models that regularize $\mathbf{b}$ or $\boldsymbol{\beta}$? (2) Which type of regularization yields better performance according to out-of-sample metrics? The performance of the eight-factor models of the SDF in cross-sectional asset pricing is the focus of this analysis. Our aim is to determine whether the crypto-eight model provides better performance.

Here, as advocated by Kozak, Nagel, and Santosh (2020), we utilize the same 13 factors as test assets. In this case, the factors not only serve as the assets whose returns need to be explained but also act as candidate factors that may become priced factors in the SDF formulation. In so doing, we consider a statistical approach that can effectively handle a wide range of potential crypto factors and assets, accurately determining their corresponding loadings in the SDF model. Then we employ the models estimated in a baseline cross-section of 13 crypto portfolios to price

the larger cross-section of 33 crypto assets without the need for additional estimation of the SDF.

Table 15 shows the out-of-sample performance comparison for models. We display the $100 \times \text{HJ}_{\text{distance}}^+$, while also showing MAE_{OLS} and cross-sectional R^2 values. In particular, we present results for an eight-factor model that distill the potential factors examined with the highest seven risk loadings along the lines of Kozak, Nagel, and Santosh (2020) — superscripted “kns” — as well as a corresponding two-factor characterization based on $\mathbb{m}_{t+\Delta} = 1 - \mathbf{b}_{\text{market}}(f_{t+\Delta}^{\text{market}} - \mu_{\text{market}}) - \mathbf{b}_{\text{mve}}(\mathbb{m}_{t+\Delta}^{\text{mve}} - \mu_{\text{mve}})$, where $\mathbb{m}_{t+\Delta}^{\text{mve}} = \mathbf{b}'\mathbf{f}_{t+\Delta}$ is constructed from the seven most pronounced risk factors.

At the center of our analysis are three conclusions. When ranking models based on their performance in relation to the $\text{HJ}_{\text{distance}}^+$ criterion, the three-factor model consistently scores lower compared to other eight-factor models. Specifically, the eight-factor model supported by elastic net and ridge had the lowest $\text{HJ}_{\text{distance}}^+$, followed by the eight-factor model based on regularizing the SDF loadings ($\text{HJ}_{\text{distance}}^{+, 8 \text{ kns}}$). This is evidenced by the $100 \times (\frac{\text{HJ}_{\text{distance}}^{+, 3 \text{ factor}}}{\text{HJ}_{\text{distance}}^{+, 8 \text{ factor}}} - 1)$ and $100 \times (\frac{\text{HJ}_{\text{distance}}^{+, 3 \text{ factor}}}{\text{HJ}_{\text{distance}}^{+, 8 \text{ kns}}} - 1)$ bootstrap computed values of 8.84 and 8.85, respectively, with the White’s two-sided p -value of 0.00. Additionally, the bootstrap p -value that tests the hypothesis that $\text{HJ}_{\text{distance}}^{+, 8 \text{ factor}} > \text{HJ}_{\text{distance}}^{+, 3 \text{ factor}}$ has a p -value of 0.14 implying that this eight-factor model does better than the three-factor model in 86% of the 10,000 bootstrap draws. We further observe that the eight-factor model does slightly better than the eight-factor model with regularized SDF loadings, as supported by $100 \times (\frac{\text{HJ}_{\text{distance}}^{+, 8 \text{ kns}}}{\text{HJ}_{\text{distance}}^{+, 8 \text{ factor}}} - 1)$ bootstrap obtained value of 0.1, with a White’s two-sided p -value of 0.08. The bootstrap p -value for $\text{HJ}_{\text{distance}}^{+, 8 \text{ factor}} > \text{HJ}_{\text{distance}}^{+, 8 \text{ kns}}$ is 0.54.

Our analysis also elicits the finding that the MAE_{OLS} values are lower for the eight-factor models, contrasting higher values for the two-factor and the three-factor models (implying worsening fit). Furthermore, scoring with other metrics such as $A^{|\alpha|}$ and $A^{\bar{R}^2}$ is also better for the two eight-factor models. Overall, our analysis is consistent with the pattern that incorporating more than three factors into the model results in better crypto pricing.

4. Interpretation of the risk factors in the SDF specifications

The essence of the ensuing analysis is to investigate the relationship between innovation in volatility variables, denoted ΔVOL_t^ℓ , and key crypto factors. Additionally, we utilize ΔVOL_t^ℓ to evaluate the pricing of ten crypto characteristic-sorted portfolios that underlie the factor construction. The premise is that investors expect to be compensated for holding underperforming crypto assets during periods of positive volatility innovation, as this indicates a high SDF.

For our analysis, the following innovations in volatility are constructed:

- $\Delta\text{VOL}_{\{t+\Delta\}}^{\text{S\&P 500}}$. Innovation in equity volatility. This volatility is constructed by analyzing the five-minute returns of E-mini S&P 500 futures (Friday-to-Friday (390 observations)).
- $\Delta\text{VOL}_{\{t+\Delta\}}^{\text{vix futures}}$. Innovation in VIX futures volatility. Weekly VIX volatility is constructed by analyzing the five-minute returns of VIX futures.
- $\Delta\text{VOL}_{\{t+\Delta\}}^{\text{bond}}$. Innovation in volatility of futures on 30-year Treasury bond. Weekly volatility is constructed by analyzing the five-minute returns of Treasury bond futures.
- $\Delta\text{VOL}_{\{t+\Delta\}}^{\text{commodity}}$. Innovation in commodity index futures volatility. Weekly volatility is constructed by analyzing the underlying five-minute returns of commodity index futures.
- $\Delta\text{VOL}_{\{t+\Delta\}}^{\text{dollar index}}$. Innovation in dollar index futures volatility. Weekly volatility is constructed by analyzing the underlying five-minute returns of dollar index futures.

We use the method of weekly time-series regression to analyze the crypto factors, including their long and short components, and their relationships with volatility innovations. We then report the most pertinent results in Table 16. The regression model is as follows:

$$f_{t+\Delta}^{\text{characteristic}} = \text{constant} + \text{slope} \times \Delta\text{VOL}_{t+\Delta}^\ell + e_{t+\Delta}. \quad (8)$$

The variable $\Delta\text{VOL}_{t+\Delta}^\ell$ represents the change in intraday measured volatility over $t - \Delta$ and $t + \Delta$.

Overall, we interpret our findings as an indication that crypto investors require compensation

for strategies that perform poorly when the investing environment deteriorates and that positive innovations in volatility are good proxies for such economic states. See, among others, Liew and Vassalou (2000), Chen and Petkova (2012), and Bakshi, Gao, and Rossi (2019).

Our results reveal that the size factor is positively associated with $\Delta\text{VOL}_{t+\Delta}^{\text{bond}}$ while the crypto momentum factor has an inverse relation with $\Delta\text{VOL}_{t+\Delta}^{\text{commodity}}$. This pattern is also evident in the regressions involving the short and long leg of size and momentum factor, respectively. In economic terms, large-cap (winner) cryptos decline when bond (commodity) volatility innovation is positive. However, the returns from going long (short) small-cap (losers) do not show a significant relationship with $\Delta\text{VOL}_{t+\Delta}^{\text{bond}}$ ($\Delta\text{VOL}_{t+\Delta}^{\text{commodity}}$). This finding is crucial because the size (momentum) factor returns are tilted towards the short (long) leg. Additionally, the lottery (downside risk) factor has a significant positive (negative) relationship with $\Delta\text{VOL}_{t+\Delta}^{\text{dollar index}}$ ($\Delta\text{VOL}_{t+\Delta}^{\text{commodity}}$).

Using the findings described above, we demonstrate that fluctuations in commodity volatility have a significant impact on the SDF when using the ten-momentum portfolios as test assets. Formalizing our analysis, we modify the specification in (1) as follows:

$$\mathbb{M}_{t+\Delta} = 1 - b_{\text{market}}(f_{t+\Delta}^{\text{market}} - \mu_{\text{market}}) - b_{\text{size}}(f_{t+\Delta}^{\text{size}} - \mu_{\text{size}}) - b_{\text{vol}}(\Delta\text{VOL}_t^{\ell} - \mu_{\text{vol}}). \quad (9)$$

We find that the SDF loading on $\Delta\text{VOL}_{t+\Delta}^{\text{commodity}}$ is significantly negative. Additionally, the risk premium associated with $\Delta\text{VOL}_{t+\Delta}^{\text{commodity}}$ is negative and statistically significant, which indicates that an increase in volatility is linked to a decrease in potential investment opportunities. These results affirm that $\Delta\text{VOL}_{t+\Delta}^{\text{commodity}}$ is able to explain the variation in crypto portfolios formed on momentum. In sum, we are able to identify a source of risk underlying crypto momentum returns.

Positive innovations in Treasury bond futures volatility, as indicated by a higher $\Delta\text{VOL}_{t+\Delta}^{\text{bond}}$, are often associated with unfavorable economic states. This aspect is captured by the size factor, which performs well when $\Delta\text{VOL}_{t+\Delta}^{\text{bond}}$ increases. This is attributable to the returns of small cap

cryptos having a significant negative coefficient, which aligns with the positive SDF loading on $\Delta\text{VOL}_{t+\Delta}^{\text{bond}}$ and the positive risk premiums associated with this factor. This analysis illustrates a source of risk underlying crypto size returns.

We gather additional evidence to show that the positive innovations in commodity and bond volatility are distinct indicators of worsening investment opportunities. We note that the $\Delta\text{VOL}_{t+\Delta}^{\text{bond}}$ variable has no significant impact on the ten momentum portfolios, while $\Delta\text{VOL}_{t+\Delta}^{\text{commodity}}$ is disconnected from the ten size portfolios.

We also investigate the impact of changes in alternative measures of return volatility, specifically relating to fluctuations in the dollar index volatility. While these fluctuations result in a positive SDF loading based on our ten lottery and ten golden cross portfolios in GMM estimations, there is no significant evidence of factor risk premiums. Furthermore, when evaluating downside risk portfolios, there is no statistically significant factor risk premium associated with commodity volatility innovation. Although there is a significant SDF loading associated with $\Delta\text{VOL}_{t+\Delta}^{\text{bond}}$ on skewness, the risk premium linked to $\Delta\text{VOL}_{t+\Delta}^{\text{bond}}$ is weaker compared to other volatility measures.

In essence, our analysis demonstrates that the Fama-MacBeth regressions support accurate model pricing when incorporating dollar index volatility as a factor and the ten lottery portfolios, as the χ_{NW}^2 p -value is 0.11. Correct pricing is also supported for the ten skewness, downside, and golden cross portfolios when incorporating $\Delta\text{VOL}_{t+\Delta}^{\text{bond}}$, $\Delta\text{VOL}_{t+\Delta}^{\text{commodity}}$, and $\Delta\text{VOL}_{t+\Delta}^{\text{dollar index}}$, respectively. However, our analysis, including the χ_{NW}^2 test, shows that the model is unable to accurately price the ten size and momentum portfolios, with a p -value less than 0.05.

One further observation from this analysis is that $\Delta\text{VOL}_{t+\Delta}^{\text{S\&P 500}}$ and $\Delta\text{VOL}_{t+\Delta}^{\text{vix futures}}$ do not significantly influence the returns of any crypto factors. Innovations in equity volatility and VIX volatility are likely driven by a host of developments (such as institutional investors, substitution between stock and bond, and geopolitical risks), and may be an inadequate proxy for changes in

the investment opportunity set for crypto investors.

In summary, our novelty is to show the dimensions of risk that affect crypto factors and portfolios. We offer an economic perspective on the crypto factors with our findings suggesting that these factors, which are used to explain crypto cross-section, may reflect information about the overall performance of the economy.

5. Conclusion

Two main aspects of our investigation are the selection of crypto test assets from various portfolios and the development of a crypto pricing model for evaluation. The selected portfolios consist of 33 diverse collections, while the pricing model consists of 13 crypto factors. This approach allows for efficient testing and development of new nested and non-nested crypto models.

Creating cross-sections of portfolios for cryptos is an essential step in understanding their underlying characteristics. However, the process of grouping cryptos based on their characteristics presents a fundamental challenge. This is because it is crucial to select a diverse set of cryptos that collectively represent the entire SDF conditioned on characteristics. The complexity of dependencies among multiple characteristics, the numerous factors and portfolios that may result, redundancy in characteristics, and the interpretability of chosen features are some of the difficulties faced in this process.

To address these obstacles, our approach focuses on creating compact and informative cross-sections of test assets based on multiple crypto characteristics. Our analysis provides support for eight-factor models for cryptos, moving beyond the relevance of the three-factor model. Our methods combine machine-learning algorithms in conjunction with traditional asset pricing approaches that estimate the SDF loading and assess cross-sectional pricing errors.

Asset pricing studies have made advancements in identifying characteristics that can explain

variations in expected returns. However, a limitation of this research is that it has only tested these models on portfolios constructed from a limited number of known return predictors. As a result, it is uncertain how well these models would perform when faced with a larger set of cross-sectional return predictors, which is often a challenge due to the high number of variables involved. Our study is part of a literature that uses machine learning techniques to overcome the challenges posed by high dimensionality in asset pricing. We specifically focus on examining cross-sectional pricing in the crypto market, both in and out-of-sample, allowing for the development of new crypto models.

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Table 1

Number, market capitalization, trading volume, and circulation rates of cryptos by year

This table describes the attributes of cryptocurrencies categorized by year. Panel A features all cryptos in our sample, whereas Panels B and C solely comprise crypto coins and tokens, respectively. Statistics for market capitalization and trading volume are presented in U.S. dollars. The circulation rate shows the percentage of cryptos actively circulating in the market. The data spans the time period from January 3, 2014, to January 26, 2024, constituting 525 weeks. This data is sourced from coinmarketcap.com.

Year	Number	Market cap. (million, \$) Mean	Trading volume (thousand, \$) Mean	Circulation rate Mean
Panel A: Cryptos (coins and tokens), January 3, 2014, to January 26, 2024 (525 weeks)				
2023	5389	39,625	16,836	0.88
2022	3494	63,626	46,796	0.82
2021	2201	8,883	126,384	0.73
2020	1274	732	96,848	0.69
2019	834	1,655	65,890	0.68
2018	624	602	28,89	0.69
2017	297	555	26,746	0.78
2016	121	89	1,079	0.94
2015	83	55	486	0.94
2014	58	131	701	0.93
Panel B: Crypto coins, January 3, 2014, to January 26, 2024 (525 weeks)				
2023	551	4,532	68,061	0.87
2022	504	2,631	162,367	0.85
2021	453	3,641	307,435	0.83
2020	419	752	169,556	0.81
2019	336	940	104,634	0.81
2018	278	1,273	50,113	0.82
2017	175	829	39,740	0.88
2016	114	94	1,132	0.94
2015	78	59	516	0.94
2014	55	138	739	0.92
Panel C: Crypto tokens, January 3, 2014, to January 26, 2024 (525 weeks)				
2023	4838	43,622	11,002	0.88
2022	2990	73,907	27,315	0.81
2021	1748	10,242	79,464	0.71
2020	855	722	61,217	0.64
2019	498	2,138	39,749	0.60
2018	346	63	11,848	0.58
2017	122	161	8,107	0.63
2016	7	11	217	0.95
2015	5	2	12	1.0
2014	3	4	4	1.0

Table 2

Properties of excess returns of cryptos (percentage per week)

This table includes data covering a period from January 3, 2014, to January 26, 2024, which comprises 525 weeks. The constructed data consists of the weekly excess returns (%) for both value-weighted and equal-weighted portfolios of cryptocurrencies, as well as the excess returns of four prominent cryptos: Bitcoin, Ethereum, Ripple, and Dogecoin. The mean, standard deviation (SD), bootstrap 90% lower and upper boundaries, Newey and West (1987) p -values (shown as NW[p]), minimum, median, maximum, and first-order autocorrelation coefficient (shown as Acf) are calculated. We also calculate the time-series fraction of returns that are positive, sample skewness, and sample excess kurtosis.

	Mean (%) (weekly)	SD (%) (weekly)	Block Bootstrap	NW[p]	Min.	Median	Max.	Acf	$1_{\{r>0\}}$ (%)	Skewness	Kurtosis (excess)
Panel A: Weekly excess returns (%) of value-weight and equal-weight portfolios											
$r_{t+\Delta}^{e,vwi}$: Coins and tokens	1.9	19.1	[0.5 2.9]	(0.01)	-65	0.4	139	-0.27	54	1.66	10.6
$r_{t+\Delta}^{e,vwi}$: Coins	1.0	11.0	[0.2 2.0]	(0.06)	-68	0.5	51	0.07	54	-0.58	6.1
$r_{t+\Delta}^{e,vwi}$: Tokens	3.1	27.0	[1.0 4.7]	(0.01)	-70	0.0	198	-0.18	49	2.37	12.3
$r_{t+\Delta}^{e,ewi}$: Coins and tokens	3.5	11.6	[2.3 4.8]	(0.00)	-38	2.5	62	0.31	63	0.71	3.2
$r_{t+\Delta}^{e,ewi}$: Coins	3.7	11.3	[2.5 5.0]	(0.00)	-37	2.9	53	0.28	65	0.60	2.3
$r_{t+\Delta}^{e,ewi}$: Tokens	4.8	19.9	[3.1 6.6]	(0.00)	-70	2.5	180	0.10	58	2.81	17.7
Panel B: Weekly excess returns (%) of Bitcoin, Ethererum, Ripple, and Dogecoin											
$r_{t+\Delta}^{e,bitcoin}$	1.2	9.8	[0.5 2.1]	(0.01)	-39	0.6	51	0.10	54	0.40	2.7
$r_{t+\Delta}^{e,ethereum}$	2.9	17.3	[1.3 4.7]	(0.00)	-45	0.3	142	0.08	43	2.69	16.1
$r_{t+\Delta}^{e,ripple}$	2.2	21.2	[0.4 4.4]	(0.06)	-52	-1.5	200	0.27	42	3.65	22.4
$r_{t+\Delta}^{e,dogecoin}$	3.0	25.8	[0.7 4.8]	(0.02)	-46	-0.7	252	0.06	46	5.73	49.3

Table 3
Construction of 13 factors in the crypto market

This table describes the underlying crypto-traded factors. In order to create a comprehensive view of cryptos, several features are taken into consideration, including macroeconomic events, fundamental analysis, volatility, market sentiment, and technical indicators. First, the macroeconomic environment, such as economic conditions and government regulations, can impact the performance of cryptos. Second, fundamental analysis, such as coskewness, market capitalization, momentum, and trading volume, are taken into account. Third, cryptos are known for their high volatility and lottery features. Additionally, market sentiment, including investor speculation, can influence the direction of crypto markets. We include the effects of speculation by using metrics like seven-day price momentum and volume. Finally, crypto markets are technical in nature, with support and resistance levels playing a significant role in price movements. We include technical indicators to capture market trends and relative demand by implementing golden cross. Factors are constructed weekly over t to $t + \Delta$. The data sample spans January 3, 2014, to January 26, 2024 (total of 525 weeks). The data source is coinmarketcap.com.

Factor	Construction of the factors in the crypto market
$f_{t+\Delta}^{\text{market}}$	<u>Crypto market factor</u> . Value-weight excess returns of all cryptos from t to $t + \Delta$.
$f_{t+\Delta}^{\text{size}}$	<u>Crypto size factor</u> . Based on market capitalization. The size factor is calculated as the returns from a long (short) position in the bottom (top) size group. The breakpoints are 30%, 40%, and 30%.
$f_{t+\Delta}^{\text{reversal}}$	<u>Crypto reversal factor</u> . Cryptos are sorted based on previous <i>four-week</i> returns. This factor is the return of a long and short position in losers and winners. The breakpoints are 30%, 40%, and 30%.
$f_{t+\Delta}^{\text{momentum}}$	<u>Crypto momentum factor</u> . Cryptos are sorted based on previous <i>seven-day</i> returns. This factor is the return from a long and short position in winners and losers. The breakpoints are 30%, 40%, and 30%.
$f_{t+\Delta}^{\text{attention}}$	<u>Crypto attention factor</u> . Cryptos are sorted based on previous seven-day volume. The factor is the return of a long and short position in low and high volume cryptos. The breakpoints are 30%, 40%, and 30%.
$f_{t+\Delta}^{\text{liquidity}}$	<u>Crypto liquidity factor</u> . Cryptos are sorted based on previous four-week volume. The factor is the return of a long and short position in low and high volume cryptos. The breakpoints are 30%, 40%, and 30%.
$f_{t+\Delta}^{\text{lottery}}$	<u>Crypto lottery factor</u> . Cryptos are sorted based on the previous 30-day volatility. The factor is the return of a long and short position in high and low volatility cryptos. The breakpoints are 30%, 40%, and 30%. For stock market evidence, see Kumar (2009) and Bali, Hirshleifer, Peng, and Tang (2019).
$f_{t+\Delta}^{\text{circulation}}$	<u>Crypto circulation factor</u> . Cryptos are sorted based on circulation rate. The factor is the return of a long and short position in low and high circulation cryptos. The breakpoints are 50%, and 50%.
$f_{t+\Delta}^{\text{skewness}}$	<u>Crypto skewness factor</u> . Cryptos are sorted based on realized skewness over the previous 30-days. The factor is the return of a long and short position in low and high skewness cryptos. The breakpoints are 30%, 40%, and 30%.
$f_{t+\Delta}^{\text{coskew}}$	<u>Crypto coskewness factor</u> . Cryptos are sorted based on realized coskewness (see Harvey and Siddique (2000) for equity market evidence) over the previous 30-days. The factor is the return of a long and short position in high and low coskewness cryptos. The breakpoints are 30%, 40%, and 30%.
$f_{t+\Delta}^{\text{downside}}$	<u>Crypto downside risk factor</u> . Cryptos are sorted based on downside risk regression (see Lettau, Maggiori, and Weber (2014) for equity market evidence) using previous 30-day threshold. The factor is the return of a long and short position in low and high downside risk cryptos. The breakpoints are 30%, 40%, and 30%.
$f_{t+\Delta}^{\text{upside}}$	<u>Crypto upside potential factor</u> . Cryptos are sorted based on upside potential regression using previous 30-day threshold. The factor is the return of a long and short position in high and low upside potential cryptos. The breakpoints are 30%, 40%, and 30%.
$f_{t+\Delta}^{\text{golden cross}}$	<u>Crypto golden cross factor</u> . Cryptos are sorted based on the differences between average returns in the past seven-days with their average returns in the past 12 weeks. The factor is the return of a long and short position in low and high categories. The breakpoints are 30%, 40%, and 30%.

Summary statistics of the 13 crypto factors (percentage per week)

Factors	Mean (%) (weekly)	SD (%) (weekly)	Block Bootstrap	NW[p]	Min.	Max.	Acf	$1_{\{r>0\}}$ (%)	Skewness	Kurtosis (excess)
$f_{t+\Delta}^{\text{market}}$	1.9	19.1	[0.5 2.9]	(0.01)	-65	139	-0.27	54	1.66	10.6
$f_{t+\Delta}^{\text{size}}$: long (short) small (large) cap	2.3	20.3	[1.2 3.9]	(0.01)	-140	85	-0.18	59	-1.0	9.5
$f_{t+\Delta}^{\text{reversal}}$: long (short) four-week decliners (risers)	1.1	37.2	[-2.0 4.2]	(0.52)	-260	238	0.01	45	0.7	13.5
$f_{t+\Delta}^{\text{momentum}}$: long (short) seven-day winners (losers)	1.6	40.1	[-0.6 4.7]	(0.36)	-231	241	-0.08	54	-0.5	10.1
$f_{t+\Delta}^{\text{attention}}$: long (short) seven-day low (high) volume	3.3	36.1	[1.3 5.8]	(0.01)	-161	249	-0.12	48	2.2	12.8
$f_{t+\Delta}^{\text{liquidity}}$: long (short) four-week low (high) volume	2.8	32.2	[0.8 4.9]	(0.02)	-172	206	-0.11	50	1.1	8.1
$f_{t+\Delta}^{\text{lottery}}$: long (short) high (low) volatility	2.5	37.5	[-0.3 4.7]	(0.11)	-78	245	-0.15	41	2.2	8.8
$f_{t+\Delta}^{\text{circulation}}$: long (short) low (high) circulation	1.5	18.9	[0.1 2.5]	(0.04)	-73	139	-0.25	54	1.6	12.0
$f_{t+\Delta}^{\text{skewness}}$: long (short) low (high) skewness	1.1	28.3	[-1.7 3.0]	(0.47)	-115	224	0.07	48	2.1	13.6
$f_{t+\Delta}^{\text{coskew}}$: long (short) high (low) coskewness	3.5	29.8	[2.0 5.5]	(0.00)	-135	220	-0.07	54	1.4	10.9
$f_{t+\Delta}^{\text{downside}}$: long (short) low (high) downside	4.2	35.0	[1.9 6.6]	(0.00)	-105	195	-0.08	52	1.1	4.7
$f_{t+\Delta}^{\text{upside}}$: long (short) high (low) upside	0.7	37.6	[-1.3 3.8]	(0.66)	-233	211	-0.04	50	0.0	9.4
$f_{t+\Delta}^{\text{golden cross}}$: long (short) low (high) golden cross	1.6	38.3	[-1.1 3.7]	(0.28)	-240	214	-0.10	49	0.5	10.1

[illegible]

Table 5

Average returns of the 33 crypto test assets (percentage per week)

Tabulated are return summary statistics of test portfolios created for the crypto market. The mean, standard deviation (SD), bootstrap 90% lower and upper boundaries, Newey and West (1987) p -values, minimum, median, maximum, and first-order autocorrelation coefficient (shown as Acf) are calculated. We also calculate the time-series fraction of returns that are positive, sample skewness, and sample excess kurtosis. The data sample spans from January 3, 2014, to January 26, 2024 (525 weeks).

Test Assets		Mean (%) (weekly)	SD (%) (weekly)	Block Bootstrap	NW[p]	Min.	Median	Max.	Acf	$1_{\{r>0\}}$ (%)	Skewness	Kurtosis (excess)
Size	Decile 1	9.78	21.8	[7.99 11.38]	(0.00)	-38	6.89	181	0.00	71	2.3	10.7
	Decile 5	3.06	16.6	[1.53 4.45]	(0.00)	-40	0.95	141	0.23	56	2.5	15.6
	Decile 10	1.92	19.2	[0.46 2.88]	(0.01)	-65	0.45	139	-0.27	54	1.7	10.6
Reversal	Decile 1	5.59	37.4	[2.21 8.75]	(0.00)	-69	-1.43	250	0.04	46	3.1	13.6
	Decile 5	0.46	15.8	[-0.86 1.76]	(0.55)	-69	0.00	140	0.04	50	2.3	19.4
	Decile 10	2.87	33.1	[0.61 5.46]	(0.06)	-70	-1.51	251	0.02	47	2.1	10.6
Momentum	Decile 1	3.66	35.0	[0.84 5.95]	(0.02)	-69	-1.58	251	-0.05	46	2.8	14.1
	Decile 5	0.52	14.1	[-0.71 1.73]	(0.47)	-63	-0.23	78	0.15	49	0.9	6.5
	Decile 10	1.44	34.9	[-0.98 3.86]	(0.34)	-70	-3.66	231	0.00	41	2.2	9.4
Attention	Decile 1	3.82	25.1	[1.81 5.76]	(0.00)	-67	0.20	173	0.03	53	2.2	9.7
	Decile 5	1.10	19.4	[-0.61 2.74]	(0.27)	-65	-0.46	203	0.15	47	2.8	25.0
	Decile 10	1.75	11.9	[0.70 2.94]	(0.01)	-40	0.67	101	0.28	56	2.2	15.0
Liquidity	Decile 1	4.84	24.0	[3.10 6.67]	(0.00)	-65	0.50	135	0.01	55	1.9	7.2
	Decile 5	0.76	20.6	[-0.90 2.55]	(0.46)	-70	-0.64	153	0.09	46	1.7	13.4
	Decile 10	2.01	14.4	[0.95 3.24]	(0.00)	-69	0.77	163	0.06	56	3.6	37.4
Lottery	Decile 1	0.70	8.9	[-0.04 1.48]	(0.11)	-37	0.15	38	0.14	52	0.2	2.6
	Decile 5	1.35	18.8	[-0.23 3.21]	(0.18)	-54	-1.06	135	0.15	46	2.2	10.8
	Decile 10	5.91	49.0	[2.28 8.86]	(0.00)	-70	-3.77	250	-0.16	44	2.3	7.5
Skewness	Decile 1	1.45	23.7	[-0.08 3.25]	(0.16)	-66	-0.16	238	-0.11	49	4.1	33.3
	Decile 5	-0.02	17.9	[-1.31 1.46]	(0.98)	-67	-0.41	143	0.03	48	2.5	21.3
	Decile 10	1.59	20.8	[-0.24 3.49]	(0.15)	-70	-0.34	120	0.17	48	1.4	7.1
Coskew	Decile 1	0.49	21.9	[-0.90 2.23]	(0.61)	-70	-0.69	185	0.06	46	2.5	19.1
	Decile 5	2.26	20.3	[0.59 3.99]	(0.02)	-54	-0.34	152	0.01	49	2.6	12.8
	Decile 10	3.03	25.3	[0.98 4.97]	(0.01)	-70	-0.96	220	0.03	46	2.8	16.2
Downside	Decile 1	5.20	38.0	[2.50 8.17]	(0.00)	-70	-2.46	248	-0.10	46	2.3	8.5
	Decile 5	1.14	15.8	[0.00 2.57]	(0.14)	-43	-0.15	198	0.09	49	5.1	55.2
	Decile 10	1.58	35.0	[-0.98 3.73]	(0.27)	-69	-1.59	237	-0.07	44	2.7	13.4
Upside	Decile 1	1.88	35.0	[-0.82 3.91]	(0.21)	-70	-2.46	211	-0.04	44	2.2	8.8
	Decile 5	0.88	13.1	[-0.23 1.81]	(0.16)	-69	0.25	126	0.05	52	1.5	18.5
	Decile 10	1.42	33.8	[-0.90 3.86]	(0.32)	-70	-2.69	248	-0.04	43	2.9	14.1
Golden Cross	Decile 1	4.06	34.7	[1.28 6.37]	(0.01)	-69	-0.74	247	-0.10	46	2.9	13.3
	Decile 5	1.55	16.3	[0.12 3.05]	(0.07)	-53	0.03	133	0.14	50	2.5	16.8
	Decile 10	0.26	31.5	[-1.83 2.97]	(0.86)	-70	-2.78	243	0.00	40	2.2	11.5

Table 6

Factors selected by the machine-learning algorithm and cross-validation

In this table, we present the results from the factor selection exercise. For a dense multifactor asset pricing model, we minimize the regularized objective associated with (4), that is, LASSO, ridge, and elastic net. We adopt a fivefold cross-validation approach to estimate the parameters of the model. Then, we select the combination, $\mathbf{f}^{\text{best}}$, that achieves the highest $R^2_{\text{elastic-net}}[\mathbf{f}^{\text{best}}]$ in equation (5). The circulation factor is never selected by the machine-learning algorithm. In the case of the three-factor model derived from elastic net, LASSO, and ridge, for example, only the β coefficients associated with market, size, and lottery are nonzero, while all other β coefficients are equal to zero.

K													
Factors	Method	$f_{t+\Delta}^{\text{market}}$	$f_{t+\Delta}^{\text{size}}$	$f_{t+\Delta}^{\text{reversal}}$	$f_{t+\Delta}^{\text{momentum}}$	$f_{t+\Delta}^{\text{attention}}$	$f_{t+\Delta}^{\text{liquidity}}$	$f_{t+\Delta}^{\text{lottery}}$	$f_{t+\Delta}^{\text{skewness}}$	$f_{t+\Delta}^{\text{coskew}}$	$f_{t+\Delta}^{\text{downside}}$	$f_{t+\Delta}^{\text{upside}}$	$f_{t+\Delta}^{\text{golden cross}}$
12	Elastic net, LASSO, ridge	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
11	Elastic net, LASSO, ridge	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓
10	Elastic net, LASSO, ridge	✓	✓	✓	✓			✓	✓	✓	✓	✓	✓
9	Elastic net, LASSO, ridge	✓	✓	✓	✓			✓	✓		✓	✓	✓
8	Elastic net, ridge	✓	✓	✓				✓	✓		✓	✓	✓
8	LASSO	✓	✓	✓	✓			✓	✓		✓	✓	
7	Elastic net, LASSO, ridge	✓	✓	✓	✓			✓			✓	✓	
6	Elastic net, LASSO, ridge	✓	✓	✓	✓			✓				✓	
5	Elastic net, LASSO, ridge	✓	✓	✓	✓			✓			✓		
4	Elastic net, LASSO, ridge	✓	✓		✓			✓					
3	Elastic net, LASSO, ridge	✓	✓					✓					
How often?		11	11	9	9	1	2	11	6	3	8	8	5

Table 7

Performance of models selected by the machine-learning algorithm with fivefold cross-validation

This table assesses the effectiveness of factor models in explaining monthly excess returns on various crypto portfolios. The 33 portfolios are classified by characteristics. We show (i) the average absolute intercept value, (ii) the ratio of average absolute alpha to the average demeaned return on portfolio, and (iii) the GRS statistic determining if the intercept estimates are significantly different from zero. The two columns for “Alphas” are based on the corresponding calculation in Fama and French (2015, Table 5). We show the average adjusted R^2 from the 33 portfolios as $A\bar{R}^2$. Additionally, we show the Hansen and Jagannathan (1997) distance with identity matrix (Cochrane (2005)).

K factors	Supporting Method	Alphas		$A\bar{R}^2$ (%)	GRS		$HJ_{\text{distance}}^+ \times 100$
		$A^{ \alpha }$ (%) (weekly)	$\frac{A^{ \alpha }}{A^{ \bar{r} }}$ (%)		stat.	p -value	
13 factors	Dense model	1.13	72.4	37.6	3.75	0.00	16.46
12 factors	Elastic net, LASSO, ridge	1.13	72.5	37.2	3.74	0.00	12.71
11 factors	Elastic net, LASSO, ridge	1.13	72.5	37.0	3.74	0.00	14.26
10 factors	Elastic net, LASSO, ridge	1.13	72.6	36.2	3.76	0.00	9.95
9 factors	Elastic net, LASSO, ridge	1.16	74.1	35.9	3.81	0.00	9.14
8 factors	Elastic net, ridge	1.11	70.9	35.4	3.87	0.00	8.88
8 factors	LASSO	1.21	77.7	35.3	3.95	0.00	9.24
7 factors	Elastic net, LASSO, ridge	1.22	78.1	34.5	3.95	0.00	9.22
6 factors	Elastic net, LASSO, ridge	1.37	87.6	32.8	4.20	0.00	9.53
5 factors	Elastic net, LASSO, ridge	1.37	87.6	31.1	4.14	0.00	9.53
4 factors	Elastic net, LASSO, ridge	1.35	86.7	29.2	4.13	0.00	9.52
3 factors	Elastic net, LASSO, ridge	1.33	85.4	26.1	4.12	0.00	9.85
3 factors	market, size, reversal	1.36	87.5	25.3	4.09	0.00	10.30
3 factors	market, size, momentum	1.40	89.8	25.5	4.10	0.00	9.92

Table 8

Cross-sectional pricing results of best models selected by the machine-learning algorithm with fivefold cross-validation

The following table displays the estimated factor risk premiums and SDF loadings using alternative models derived from the analysis in Table 7. The results are based on an eight-factor model, and we also present results from an expanded nine-factor model and a restricted seven-factor model. The estimations are conducted using a two-step GMM method on a dataset over the sample period of January 3, 2014, to January 26, 2024 (covering 525 weeks). The level of significance is superscripted by ** if the p -value is lower than 0.05, and * if the p -value is between 0.05 and 0.10. χ^2_{NW} represents the test of whether all pricing errors are zero (Cochrane (2005, 12.2, page 237)) and is distributed χ^2_{N-K} , where K is the number of factors. We determine the goodness-of-fit using both the uncentered R^2 and the MAE_{OLS} as metrics.

Panel A: Model estimation results										
Method	Eight-factor models				Nine-factor model		Seven-factor model		Three-factor model	
	Elastic net, ridge		LASSO		Elastic net, LASSO, ridge		Elastic net, LASSO, ridge		market, size, momentum	
	SDF Loading	Factor Premium	SDF Loading	Factor Premium	SDF Loading	Factor Premium	SDF Loading	Factor Premium	SDF Loading	Factor Premium
	b	λ $\times 100$	b	λ $\times 100$	b	λ $\times 100$	b	λ $\times 100$	b	λ $\times 100$
$f_{t+\Delta}^{\text{market}}$	2.11**	6.96**	2.05**	7.22**	1.96**	6.93**	2.09**	6.71**	2.67**	6.49**
$f_{t+\Delta}^{\text{size}}$	3.03**	6.46**	2.90**	6.29**	3.10**	6.46**	2.95**	6.28**	2.54**	5.39**
$f_{t+\Delta}^{\text{reversal}}$	-0.40	-2.46	-0.16	-1.85	-0.52*	-2.47	0.07	0.03		
$f_{t+\Delta}^{\text{momentum}}$			0.02	-1.43	0.77**	0.20	-0.07	-1.55	-0.20	0.63
$f_{t+\Delta}^{\text{lottery}}$	0.38**	4.11**	0.38**	3.72*	0.39**	4.13**	0.38**	3.98**		
$f_{t+\Delta}^{\text{skewness}}$	0.57	3.82*	0.48	3.28	0.81**	3.85*				
$f_{t+\Delta}^{\text{downside}}$	0.46**	3.44*	0.41**	3.51*	0.39**	3.44*	0.40**	3.44*		
$f_{t+\Delta}^{\text{upside}}$	-0.01	3.29	-0.10	3.21	-0.02	3.28	-0.14	3.01		
$f_{t+\Delta}^{\text{golden cross}}$	0.22	2.83			0.94**	2.90				
Hansen's J	131 (0.00)		129 (0.00)		125 (0.00)		132 (0.00)		140 (0.00)	
{df}	{25}		{25}		{24}		{26}		{30}	
Panel B: Cross-sectional pricing errors										
R_{GLS}^2 (%)	78.5		78.4		78.2		78.1		75.6	
R_{OLS}^2 (%)	80.1		79.6		80.1		78.8		75.6	
MAE_{OLS} (%)	0.970		0.984		0.969		0.982		1.13	
χ^2_{NW}	110.2 (0.00)		307.1 (0.00)		131.4 (0.00)		77.0 (0.00)		157.5 (0.00)	
Panel C: Bootstrap evidence on model comparisons										
$100 \times \left(\frac{HJ_{\text{distance}}^{+, 3 \text{ factor}}}{HJ_{\text{distance}}^{+, 8 \text{ factor}}} - 1 \right)$ is 8.0 with White's two-sided p -value of 0.00. The bootstrap p -value is 0.14 for $HJ_{\text{distance}}^{+, 8 \text{ factor}} > HJ_{\text{distance}}^{+, 3 \text{ factor}}$.										
$100 \times \left(\frac{HJ_{\text{distance}}^{+, 3 \text{ factor}}}{HJ_{\text{distance}}^{+, 9 \text{ factor}}} - 1 \right)$ is 6.3 with White's two-sided p -value of 0.00. The bootstrap p -value is 0.24 for $HJ_{\text{distance}}^{+, 9 \text{ factor}} > HJ_{\text{distance}}^{+, 3 \text{ factor}}$.										
$100 \times \left(\frac{HJ_{\text{distance}}^{3 \text{ factor}}}{HJ_{\text{distance}}^{+, 7 \text{ factor}}} - 1 \right)$ is 5.8 with White's two-sided p -value of 0.00. The bootstrap p -value is 0.22 for $HJ_{\text{distance}}^{+, 7 \text{ factor}} > HJ_{\text{distance}}^{+, 3 \text{ factor}}$.										

Table 9

Exclusion tests: Restricting the SDF loadings on certain factors to be zero

The results presented here are obtained using a two-step GMM estimation procedure with an optimally weighted matrix. We assess the validity of four exclusion restrictions on the SDF and reject those with p -values below 0.05, indicating that the excluded factors have loadings on the SDF that are significantly different from zero. Table 8 contains the GMM results for the eight-, nine-, and seven-factor models.

	Original Model	Excluded factors in SDF	χ^2	{df}	p -value
1	Eight-factor (Elastic net, ridge)	$b_{\text{lottery}} = b_{\text{skewness}} = b_{\text{downside}} = b_{\text{upside}} = b_{\text{golden cross}} = 0$	17.7	{5}	(0.000)
2	Eight-factor (LASSO)	$b_{\text{reversal}} = b_{\text{lottery}} = b_{\text{skewness}} = b_{\text{downside}} = b_{\text{upside}} = 0$	19.3	{5}	(0.000)
3	Nine-factor	$b_{\text{reversal}} = b_{\text{lottery}} = b_{\text{skewness}} = b_{\text{downside}} = b_{\text{upside}} = b_{\text{golden cross}} = 0$	23.2	{6}	(0.000)
4	Seven-factor	$b_{\text{reversal}} = b_{\text{lottery}} = b_{\text{downside}} = b_{\text{upside}} = 0$	17.5	{4}	(0.000)

Table 10

Expected return pricing errors across the 33 crypto portfolios

To calculate the model-based average returns of various crypto portfolios, we use equation (3) to obtain the fitted average returns. The mean absolute errors (denoted MAE_{OLS}) are then computed as $(1/33)\sum_{i=1}^{33} |\text{Realized}_i - \text{Fitted}_i|$ (using the identity matrix in the calculation of fitted returns). This is reflected as weekly percentage unit.

Test Asset		Average Return (%) (weekly) (realized)	Eight-factors		Three-factors	
			Fitted	Model	Fitted	Model
			Return	Error	Return	Error
			(%)	(%)	(%)	(%)
(weekly) (weekly)						
$MAE_{OLS}^{\text{three-factor}} - MAE_{OLS}^{\text{eight-factor}} = 0.16\%$ with a one-sided p -value = 0.058						
1	Size: Decile 1	9.78	4.75	5.03	4.42	5.36
2	Size: Decile 5	3.06	3.51	-0.45	3.41	-0.36
3	Size: Decile 10	1.92	1.73	0.19	1.93	-0.01
4	Reversal: Decile 1	5.59	3.42	2.17	3.71	1.88
5	Reversal: Decile 5	0.46	2.71	-2.25	2.20	-1.74
6	Reversal: Decile 10	2.87	2.85	0.03	1.88	0.99
7	Momentum: Decile 1	3.66	3.36	0.30	3.06	0.60
8	Momentum: Decile 5	0.52	1.55	-1.03	1.60	-1.08
9	Momentum: Decile 10	1.44	1.28	0.17	1.04	0.41
10	Attention: Decile 1	3.82	4.19	-0.37	3.56	0.26
11	Attention: Decile 5	1.10	1.37	-0.27	1.37	-0.27
12	Attention: Decile 10	1.75	0.45	1.30	0.40	1.35
13	Liquidity: Decile 1	4.84	3.81	1.03	3.69	1.15
14	Liquidity: Decile 5	0.76	3.35	-2.59	3.36	-2.60
15	Liquidity: Decile 10	2.01	0.56	1.45	0.81	1.20
16	Lottery: Decile 1	0.70	-0.61	1.31	-0.43	1.13
17	Lottery: Decile 5	1.35	2.91	-1.56	2.64	-1.29
18	Lottery: Decile 10	5.91	5.30	0.61	5.51	0.39
19	Skewness: Decile 1	1.45	2.42	-0.96	1.98	-0.52
20	Skewness: Decile 5	-0.02	1.17	-1.20	1.59	-1.61
21	Skewness: Decile 10	1.59	1.33	0.25	1.93	-0.34
22	Coskew: Decile 1	0.49	1.04	-0.55	1.92	-1.43
23	Coskew: Decile 5	2.26	2.42	-0.17	2.17	0.08
24	Coskew: Decile 10	3.03	1.90	1.13	1.67	1.36
25	Downside: Decile 1	5.20	5.21	-0.01	3.59	1.62
26	Downside: Decile 5	1.14	0.86	0.29	1.01	0.14
27	Downside: Decile 10	1.58	1.31	0.27	3.15	-1.56
28	Upside: Decile 1	1.88	3.61	-1.73	5.16	-3.29
29	Upside: Decile 5	0.88	0.60	0.28	0.97	-0.10
30	Upside: Decile 10	1.42	3.41	-1.99	2.59	-1.17
31	Golden Cross: Decile 1	4.06	4.42	-0.36	3.67	0.39
32	Golden Cross: Decile 5	1.55	1.35	0.20	1.17	0.38
33	Golden Cross: Decile 10	0.26	0.77	-0.51	1.59	-1.33
Mean		2.49	2.39		2.34	
Standard deviation		2.07	1.51		1.36	

Table 11

Cross-sectional pricing results for the best models selected by the machine-learning algorithm and removing the insignificant SDF loadings in Table 8

The table displays the estimated factor risk premiums and SDF loadings for alternative models. These models are based on the analysis in Table 7 and include the restricted four- and eight-factor models. These results are based on a two-step GMM estimation procedure. The level of significance is superscripted by ** if the p -value is lower than 0.05, and * if the p -value is between 0.05 and 0.10. χ^2_{NW} represents the test of whether all pricing errors are zero (Cochrane (2005, 12.2, page 237)) and is distributed χ^2_{N-K} , where K is the number of factors.

	Panel A: Eight-factor model restricted to four factors		Panel B: Nine-factor model restricted to eight factors	
	SDF Loading b	Factor Premium λ $\times 100$	SDF Loading b	Factor Premium λ $\times 100$
$f_{t+\Delta}^{\text{market}}$	2.11**	4.5**	1.99**	5.2**
$f_{t+\Delta}^{\text{size}}$	2.93**	4.5**	3.13**	5.7**
$f_{t+\Delta}^{\text{reversal}}$	$\equiv 0$	$\equiv 0$	-0.52*	-0.7
$f_{t+\Delta}^{\text{momentum}}$			0.79**	0.8
$f_{t+\Delta}^{\text{lottery}}$	0.38**	5.4**	0.34*	4.8**
$f_{t+\Delta}^{\text{skewness}}$	$\equiv 0$	$\equiv 0$	0.84**	2.0
$f_{t+\Delta}^{\text{downside}}$	0.42**	2.6	0.40**	3.2
$f_{t+\Delta}^{\text{upside}}$	$\equiv 0$	$\equiv 0$	$\equiv 0$	$\equiv 0$
$f_{t+\Delta}^{\text{golden cross}}$	$\equiv 0$	$\equiv 0$	1.00**	3.0
Hansen's J-statistic	131 (0.00)		127 (0.00)	
{df}	{29}		{25}	
Cross-Sectional Pricing Errors				
$100 \times \text{HJ}_{\text{distance}}^+$	9.60		9.13	
R_{GLS}^2 (%)	73.4		78.2	
R_{OLS}^2 (%)	71.4		77.4	
MAE_{OLS} (%)	1.179		0.966	
χ^2_{NW}	119.6 (0.00)		141.0 (0.00)	
$A^{ \alpha }$ (%)	1.24		1.16	
$\frac{A^{ \alpha }}{A^{ \beta }}$ (%)	79.8		74.2	
$A^{\overline{R}^2}$ (%)	27.9		34.4	

Table 12

Time-series regressions of the eight-factor (crypto-eight) model and alpha tests

The eight-factor model is supported by elastic net and ridge regression results and our diagnostics. We conduct a time-series regression using this eight-factor model in (following equation (6)) and present the regression coefficient estimates. The level of significance is superscripted by ** if the p -value is lower than 0.05, and * if the p -value is between 0.05 and 0.10. The p -values are calculated using the Newey and West (1987) method, with the number of lags automatically selected. To test the hypothesis that all intercepts are equal to zero, a GMM approach was adopted; that is, $E[(\mathbf{r}_{t+\Delta}^e - \boldsymbol{\alpha} - \beta \mathbf{f}_{t+\Delta}') \otimes [\mathbf{1} \mathbf{f}_{t+\Delta}']'] = \mathbf{0}_{(297 \times 1)}$ (see Cochrane (2005, pages 233 and 234)). The Wald test has an asymptotically chi-square distribution with 33 degrees of freedom.

	Portfolio	α	β_{market}	β_{size}	β_{reversal}	β_{lottery}	β_{skewness}	β_{downside}	β_{upside}	$\beta_{\text{golden cross}}$	$\bar{R}^2$ (%)
Joint test of $\boldsymbol{\alpha} = \mathbf{0}$ for <u>eight-factor</u> model: $\chi^2(33) = 183$, p -value=0.000, # of α_i significant: 13 out of 33											
Joint test of $\boldsymbol{\alpha} = \mathbf{0}$ for <u>three-factor</u> model: $\chi^2(33) = 178$, p -value=0.000, # of α_i significant: 17 out of 33											
Size	Decile 1	6.7**	0.77**	0.72**	0.004	0.02	-0.03	0.01	-0.01	-0.02	23
	Decile 5	0.4	0.70**	0.61**	0.01	0.02	-0.03	-0.01	-0.01	-0.02	31
	Decile 10	0.03**	0.995**	-0.01**	0.00	-0.001*	0.001	0.00	0.00	0.00	100
Reversal	Decile 1	2.2*	0.78**	0.33**	0.35**	0.28**	0.20**	-0.04	0.02	0.004	48
	Decile 5	-2.0**	0.67**	0.48**	0.01	-0.04**	0.03	0.03	0.01	0.02	27
	Decile 10	0.3	0.53**	0.31**	-0.35**	0.33**	-0.20**	0.12**	0.12**	0.02	46
Momentum	Decile 1	0.3	0.51**	0.30**	0.01	0.31**	0.14*	0.02	0.01	0.41**	49
	Decile 5	-1.5**	0.63**	0.42**	0.03*	-0.04**	-0.04	-0.02	0.03*	-0.02	30
	Decile 10	-0.6	0.63**	0.21	-0.13*	0.21**	-0.13	0.09	0.08	-0.23**	28
Attention	Decile 1	1.0	0.69**	0.62**	-0.01	0.03	0.09	-0.001	0.03	-0.01	14
	Decile 5	-0.6	0.61**	0.39**	-0.05	-0.04*	0.09	-0.04*	-0.05	-0.08	20
	Decile 10	0.2	0.58**	0.28**	-0.02	-0.06**	-0.04	0.01	0.04*	-0.02	42
Liquidity	Decile 1	1.9**	0.66**	0.68**	0.03	0.06	-0.01	0.001	-0.01	-0.02	16
	Decile 5	-1.9**	0.73**	0.54**	0.03	0.03	-0.04	-0.01	0.02	0.01	21
	Decile 10	0.5	0.66**	0.25**	0.02	-0.06**	-0.09**	-0.04	0.07**	0.01	42
Lottery	Decile 1	-0.5*	0.46**	0.24**	-0.001	-0.05**	0.01	-0.01	0.02*	-0.03*	41
	Decile 5	-1.2*	0.73**	0.51**	0.02	-0.03	-0.10**	0.04	0.02	0.02	25
	Decile 10	1.6	0.64**	0.30**	0.02	0.85**	0.08	0.04	0.06	0.01	58
Skewness	Decile 1	-1.2*	0.58**	0.41**	0.18*	0.09**	0.22**	-0.03	0.07	-0.001	41
	Decile 5	-2.2**	0.62**	0.41**	0.13**	0.004	-0.10**	0.003	0.01	0.01	23
	Decile 10	-0.5	0.68**	0.36**	-0.01	0.09*	-0.24**	0.01	0.01	-0.04	33
Coskew	Decile 1	-1.34*	0.53**	0.45**	0.04	0.11**	-0.06	-0.11**	-0.08*	0.01	17
	Decile 5	-0.1	0.60**	0.50**	0.02	0.02	0.002	0.001	0.03	0.03	15
	Decile 10	-0.1	0.67**	0.34**	0.12**	0.04	-0.24**	0.21**	0.10**	0.08*	33
Downside	Decile 1	-0.2	0.58**	0.45**	-0.01	0.37**	-0.01	0.56**	0.003	-0.002	50
	Decile 5	-0.6	0.58**	0.34**	0.05	-0.04**	0.06	-0.03*	-0.01	-0.01	23
	Decile 10	0.5	0.72**	0.32**	0.01	0.31**	-0.16**	-0.41**	0.05	-0.01	36
Upside	Decile 1	-1.7	0.85**	0.51**	0.17**	0.28**	-0.04	0.03	-0.39**	0.05	44
	Decile 5	-0.9	0.53**	0.37**	0.02	0.02	-0.06**	-0.02	0.00	-0.03**	29
	Decile 10	-1.6*	0.63**	0.29**	0.10*	0.31**	0.003	0.01	0.44**	-0.06	53
Golden Cross	Decile 1	0.5	0.73**	0.30**	0.02	0.23**	0.11*	0.02	-0.03	0.45**	57
	Decile 5	-0.6	0.57**	0.41**	0.03	-0.04**	-0.05	0.04	0.05**	0.03	20
	Decile 10	-1.5	0.63**	0.27**	-0.06	0.18*	-0.09	0.03	0.04	-0.34**	33

Table 13

Factor spanning tests: Time-series regressions with the eight-factor (crypto-eight) model

This table provides the intercept and slope coefficients estimated from time-series regressions of each of the 13 crypto factors using the eight-factor model as the covariate. The level of significance is superscripted by ** if the p -value is lower than 0.05, and * if the p -value is between 0.05 and 0.10. The last column displays the adjusted R^2 ($\bar{R}^2$).

	Intercept (%)	β_{market}	β_{size}	β_{reversal}	β_{lottery}	β_{skewness}	β_{downside}	β_{upside}	$\beta_{\text{golden cross}}$	$\bar{R}^2$ (%)
$f_{t+\Delta}^{\text{market}}$	2.95**	$\equiv 0$	-0.62**	-0.02	0.11**	0.02	0.03	0.01	-0.01	59
$f_{t+\Delta}^{\text{size}}$	3.93**	-0.75**	$\equiv 0$	-0.01	-0.02	-0.02	-0.02	0.00	-0.04**	56
$f_{t+\Delta}^{\text{reversal}}$	0.32	-0.11	-0.03	$\equiv 0$	-0.02	0.66	-0.02	0.17	0.26	37
$f_{t+\Delta}^{\text{momentum}}$	1.80**	0.22	0.05	-0.08	-0.01	-0.22*	0.15	0.11	-0.70**	60
$f_{t+\Delta}^{\text{attention}}$	0.92	0.15	0.31**	0.16	0.43**	-0.06	0.11	-0.11	-0.07	21
$f_{t+\Delta}^{\text{liquidity}}$	0.66	0.24	0.32**	0.03	0.33**	-0.02	0.08	-0.11	-0.06	17
$f_{t+\Delta}^{\text{lottery}}$	0.89	0.76**	-0.12	-0.03	$\equiv 0$	0.23	0.01	0.07	0.08	25
$f_{t+\Delta}^{\text{circulation}}$	0.76	0.47**	0.06	-0.05	-0.04	0.06	-0.04	0.00	-0.001	16
$f_{t+\Delta}^{\text{skewness}}$	0.68	0.06	-0.06	0.40**	0.11*	$\equiv 0$	-0.06	-0.14**	-0.004	34
$f_{t+\Delta}^{\text{coskew}}$	1.80**	-0.08	-0.09	0.17**	0.04	0.01	0.38**	0.40**	-0.07	47
$f_{t+\Delta}^{\text{downside}}$	4.31**	0.21	-0.15	-0.03	0.02	-0.13	$\equiv 0$	-0.09	-0.01	4
$f_{t+\Delta}^{\text{upside}}$	1.06	0.05	0.00	0.26**	0.09	-0.36**	-0.10	$\equiv 0$	-0.10	6
$f_{t+\Delta}^{\text{golden cross}}$	1.84	-0.08	-0.26**	0.36**	0.09	-0.01	-0.01	-0.09	$\equiv 0$	15

Table 14

Estimated SDF loadings and factor risk premiums with (i) market and 12 principal components, (ii) dense factor model of 13 principal components, and (iii) market plus mean-variance efficient portfolio

This table presents the estimated SDF loadings and factor risk premiums using a dense factor model with 13 principal components. These principal components are derived from either (i) 12 factors orthogonalized with respect to the crypto market or (ii) all 13 factors, and the estimation is conducted using a two-step and iterated GMM method (see equation (2)) with 33 test assets. The dataset covers a sample period from January 3, 2014, to January 26, 2024, which includes 525 weeks. The significance level is denoted with ** if the p -value is less than 0.05, and * if the p -value falls between 0.05 and 0.10. χ^2_{NW} represents the test of whether all pricing errors are zero (Cochrane (2005, 12.2, page 237)) and is distributed χ^2_{N-K} , where K is the number of factors.

	Panel A: Crypto market and 12 principal components (33 assets and 13 factors)		Panel B: 13 crypto principal components (33 assets and 13 factors)			Panel C: Crypto market plus MVE portfolio (33 assets and two factors)	
	SDF Loading	Factor Premium	SDF Loading		Factor Premium	SDF Loading	
	Two Step	Fama McBeth	Two Step	Iterated GMM	Fama McBeth	Two Step	
$f_{t+\Delta}^{\text{market}}$	0.12	0.06**	$f_{t+\Delta}^{\text{pc1}}$	0.02	0.04	0.09	$f_{t+\Delta}^{\text{market}}$ 0.87**
$f_{t+\Delta}^{\text{pc1}}$	0.03	-0.21**	$f_{t+\Delta}^{\text{pc2}}$	0.01	0.04	0.11	$\Pi_{t+\Delta}^{\text{mve}}$ 3.37**
$f_{t+\Delta}^{\text{pc2}}$	0.06	-0.03	$f_{t+\Delta}^{\text{pc3}}$	-0.04	0.02	0.25*	
$f_{t+\Delta}^{\text{pc3}}$	0.08	0.08	$f_{t+\Delta}^{\text{pc4}}$	0.13	0.11	0.10	
$f_{t+\Delta}^{\text{pc4}}$	0.06	-0.40**	$f_{t+\Delta}^{\text{pc5}}$	0.20**	0.24**	-0.02	
$f_{t+\Delta}^{\text{pc5}}$	0.06	0.23**	$f_{t+\Delta}^{\text{pc6}}$	0.004	0.04	0.07	
$f_{t+\Delta}^{\text{pc6}}$	-0.31**	-0.57**	$f_{t+\Delta}^{\text{pc7}}$	0.14	0.28**	0.30**	
$f_{t+\Delta}^{\text{pc7}}$	0.08	0.26**	$f_{t+\Delta}^{\text{pc8}}$	0.08	0.34**	0.21**	
$f_{t+\Delta}^{\text{pc8}}$	-0.22	-0.47**	$f_{t+\Delta}^{\text{pc9}}$	-0.13	-0.06	0.01	
$f_{t+\Delta}^{\text{pc9}}$	-0.15	-0.03	$f_{t+\Delta}^{\text{pc10}}$	-0.48	-0.57*	-0.03	
$f_{t+\Delta}^{\text{pc10}}$	-0.69**	-0.20**	$f_{t+\Delta}^{\text{pc11}}$	0.27	0.02	-0.07	
$f_{t+\Delta}^{\text{pc11}}$	0.14	-0.14**	$f_{t+\Delta}^{\text{pc12}}$	0.69**	1.27**	0.48**	
$f_{t+\Delta}^{\text{pc12}}$	0.06	-0.10	$f_{t+\Delta}^{\text{pc13}}$	0.38	-0.04	0.08	
Hansen's J	53.8 (0.00)			53.8 (0.00)	82.9 (0.00)		
{df}	{20}			{20}	{20}		
$100 \times \text{HJ}_{\text{distance}}^+$	14.84		14.90	11.73		9.72	
R_{GLS}^2 (%)	82.3			82.3		68.5	
R_{OLS}^2 (%)	83.7			83.7		68.7	
MAE_{OLS} (%)	0.848			0.848		1.19	
χ_{NW}^2	87.8 (0.00)			94.1 (0.00)			

Table 15

Out-of-sample cross-sectional performance of models (model estimated on 13 assets same as factors but applied to 33 test portfolios)

This table assesses the effectiveness of factor models in explaining monthly excess returns on various crypto portfolios not used in estimation. The 33 portfolios considered are classified by characteristics as portfolios 1, 5, and 10. We show (i) the Hansen and Jagannathan (1997) distance with identity matrix (Cochrane (2005)), (ii) the average absolute alpha value, (iii) the ratio of average absolute alpha to the average demeaned return on portfolio, (iv) the average adjusted R^2 from the 33 portfolios as $A\bar{R}^2$, and (v) the GRS statistic. The “kns” model refers to the method of regularizing SDF loadings.

	$HJ_{\text{distance}}^+ \times 100$	$A^{ \alpha } (\%)$	$\frac{A^{ \alpha }}{A^{ \bar{r} }}$	$A\bar{R}^2 (\%)$	$R_{\text{OLS}}^2 (\%)$	$R_{\text{GLS}}^2 (\%)$	$MAE_{\text{OLS}} (\%)$	GRS
Panel A: Models estimated using 13 assets same as 13 factors and applied to 33 crypto portfolios								
Eight factors: Elastic net and ridge	9.54	1.11	0.71	35.4	80.1	78.5	0.970	3.87
Eight factors: regularizing SDF loadings	9.57	1.12	0.72	30.6	79.7	77.0	0.962	3.70
Market plus MVE: regularizing SDF loadings	10.89	1.16	0.75	18.1	68.7	68.5	1.185	3.98
Three-factors: Market, size, momentum	10.70	1.40	0.90	26.1	75.6	75.6	1.133	4.10
$100 \times \left(\frac{HJ_{\text{distance}}^+, 3 \text{ factor}}{HJ_{\text{distance}}^+, 8 \text{ factor}} - 1 \right)$ is 8.8 with White's two-sided p -value of 0.00. The bootstrap p -value is 0.14 for $HJ_{\text{distance}}^+, 8 \text{ factor} > HJ_{\text{distance}}^+, 3 \text{ factor}$.								
$100 \times \left(\frac{HJ_{\text{distance}}^+, 3 \text{ factor}}{HJ_{\text{distance}}^+, 8 \text{ kns}} - 1 \right)$ is 8.8 with White's two-sided p -value of 0.00. The bootstrap p -value is 0.13 for $HJ_{\text{distance}}^+, 8 \text{ kns} > HJ_{\text{distance}}^+, 3 \text{ factor}$.								
$100 \times \left(\frac{HJ_{\text{distance}}^+, 3 \text{ factor}}{HJ_{\text{distance}}^+, 2 \text{ kns}} - 1 \right)$ is 2.2 with White's two-sided p -value of 0.00. The bootstrap p -value is 0.44 for $HJ_{\text{distance}}^+, 2 \text{ kns} > HJ_{\text{distance}}^+, 3 \text{ factor}$.								
$100 \times \left(\frac{HJ_{\text{distance}}^+, 8 \text{ kns}}{HJ_{\text{distance}}^+, 8 \text{ factor}} - 1 \right)$ is 0.1 with White's two-sided p -value of 0.08. The bootstrap p -value is 0.54 for $HJ_{\text{distance}}^+, 8 \text{ kns} > HJ_{\text{distance}}^+, 8 \text{ factor}$.								
$100 \times \left(\frac{HJ_{\text{distance}}^+, 2 \text{ kns}}{HJ_{\text{distance}}^+, 8 \text{ factor}} - 1 \right)$ is 8.3 with White's two-sided p -value of 0.00. The bootstrap p -value is 0.31 for $HJ_{\text{distance}}^+, 8 \text{ factor} > HJ_{\text{distance}}^+, 2 \text{ kns}$.								
Panel B: Insample model estimation results								
	Three Factor	Eight Factor	Eight Factor (kns)	Two Factor (kns)				
$f_{t+\Delta}^{\text{market}}$	2.30**	2.05**	1.81**	0.62**				
$f_{t+\Delta}^{\text{size}}$	2.04**	2.25**	2.17**					
$f_{t+\Delta}^{\text{coskew}}$			0.38**					
$f_{t+\Delta}^{\text{circulation}}$			0.37					
$f_{t+\Delta}^{\text{downside}}$		0.41**	0.23*					
$f_{t+\Delta}^{\text{golden cross}}$		0.10	0.20*					
$f_{t+\Delta}^{\text{attention}}$			0.13					
$f_{t+\Delta}^{\text{liquidity}}$			0.12					
$f_{t+\Delta}^{\text{reversal}}$		0.12						
$f_{t+\Delta}^{\text{momentum}}$	0.02							
$f_{t+\Delta}^{\text{lottery}}$		0.11						
$f_{t+\Delta}^{\text{skewness}}$		0.05						
$f_{t+\Delta}^{\text{upside}}$		0.07						
$m_{t+\Delta}^{\text{mve}}$				2.52**				
$R_{\text{OLS}}^2 (\%)$	83.2	97.1	99.2	94.9				
$R_{\text{GLS}}^2 (\%)$	81.1	96.8	98.8	92.1				
$MAE_{\text{OLS}} (\%)$	0.82	0.30	0.17	0.45				
Hansen's J-statistic	21 (0.02)	8 (0.14)	2 (0.80)	24 (0.01)				
{df}	{10}	{5}	{5}	{11}				
$100 \times HJ_{\text{distance}}^+$	6.72	2.64	1.55	3.06				
$A^{ \alpha } (\%)$	1.48	0.46	0.20	1.20				
$A\bar{R}^2 (\%)$	33.9	73.8	73.7	26.1				

Table 16

Interpretation of crypto factors and innovations in volatilities of S&P 500, VIX, bond, commodity, and dollar index

For each crypto factor, we perform a regression on six innovations in volatility constructed from (i) E-mini S&P 500 futures, (ii) VIX futures, (iii) commodity index futures, (iv) futures on 30-year Treasury bond, and (v) futures on dollar index. The source of this intraday data sampled at five-minute is Barchart. Innovations in equity futures volatility and innovations in VIX futures volatility are never significant. For each crypto factor, there are two panels. The first panel presents the results of a univariate regression model that examines the relationship between the crypto factor and a innovation in a volatility (constructed from five-minute intraday data), as follows:

$$f_{t+\Delta}^{\text{characteristic}} = \text{constant} + \text{slope} \times \Delta \text{VOL}_{t+\Delta}^{\ell} + e_{t+\Delta} \quad \text{for } \ell = \overbrace{\{\text{S\&P 500, vix, commodity, bond, dollar index}\}}^{\text{futures underlier}}.$$

The significance is represented by the Newey and West p -value with a ** (p -value lower than 0.05) or * (p -value higher than 0.05 but less than 0.1). The long and short legs of carry are consistent with the construction in Table 4. In the adjacent panel, the Fama and MacBeth procedure is utilized, with p -values calculated using the Newey and West procedure. The SDF loadings are estimated by GMM, as follows:

$$E \begin{bmatrix} \underbrace{(1 - \mathbf{b}'(\mathbf{f}_{t+\Delta} - \boldsymbol{\mu}))}_{\text{market plus size plus } \Delta \text{VOL}} \otimes \underbrace{\mathbf{r}_{t+\Delta}^e}_{\text{ten crypto assets}} \\ \mathbf{f}_{t+\Delta} - \boldsymbol{\mu} \\ \text{vec}((\mathbf{f}_{t+\Delta} - \boldsymbol{\mu})(\mathbf{f}_{t+\Delta} - \boldsymbol{\mu})') - \text{vec}(\Omega_{\mathbf{f}}) \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ 0 \\ 0 \end{bmatrix}_{(19 \times 1)},$$

χ_{NW}^2 represents the test of whether all pricing errors are zero (Cochrane (2005, 12.2, page 237)) and is distributed χ_{N-K}^2 , where K is the number of factors.

Dependent variable	Univariate regression			GMM estimations and Fama-McBeth procedure						
	constant	slope	$\bar{R}^2$ (%)	Test Assets	SDF Loading b	Factor Premium λ $\times 100$	SDF Loading b	Factor Premium λ $\times 100$	SDF Loading b	Factor Premium λ $\times 100$
	$\Delta \text{VOL}_{\text{bond}}$				Crypto market		Crypto size		$\Delta \text{VOL}_{\text{bond}}$	
$f_{t+\Delta}^{\text{size}}$	0.02**	3.83*	0.15	Ten size	0.346	-2.34**	$\equiv 0$	$\equiv 0$	413.5**	0.79**
Long small cap	0.04**	0.31	-0.19				• χ_{NW}^2 p -value=0.00			
Short large cap	0.02**	-3.52	0.13							
	$\Delta \text{VOL}_{\text{commodity}}$				Crypto market		Crypto size		$\Delta \text{VOL}_{\text{commodity}}$	
$f_{t+\Delta}^{\text{momentum}}$	0.02	-1.89*	0.02	Ten momentum	0.552	6.59**	-0.165	-2.84	-131.7**	-1.02*
Long winners	0.04**	-1.41	0.05				• χ_{NW}^2 p -value=0.02			
Short losers	0.02*	0.48	-0.17							
	$\Delta \text{VOL}_{\text{dollar index}}$				Crypto market		Crypto size		$\Delta \text{VOL}_{\text{dollar index}}$	
$f_{t+\Delta}^{\text{lottery}}$	0.03	15.34**	0.44	Ten lottery	0.897**	4.67**	0.595	-4.43*	342.5**	0.01
Long high variance	0.04**	12.61**	0.21				• χ_{NW}^2 p -value=0.11			
Short low variance	0.01**	-2.73	0.11							
	$\Delta \text{VOL}_{\text{bond}}$				Crypto market		Crypto size		$\Delta \text{VOL}_{\text{bond}}$	
$f_{t+\Delta}^{\text{skewness}}$	0.01	-7.04*	0.40	Ten skewness	0.962*	-0.62	0.681	0.76	250.5**	0.08
Long low skewness	0.02**	-7.73**	0.77				• χ_{NW}^2 p -value=0.19			
Short high skewness	0.01	-0.69	-0.18							
	$\Delta \text{VOL}_{\text{commodity}}$				Crypto market		Crypto size		$\Delta \text{VOL}_{\text{commodity}}$	
$f_{t+\Delta}^{\text{downside}}$	0.04**	-2.07**	0.14	Ten downside	0.864	4.96	-1.021	-9.21*	7.769	-0.62
Long low downside	0.04**	-1.89*	0.20				• χ_{NW}^2 p -value=0.43			
Short high downside	0.00	0.18	-0.19							
	$\Delta \text{VOL}_{\text{dollar index}}$				Crypto market		Crypto size		$\Delta \text{VOL}_{\text{dollar index}}$	
$f_{t+\Delta}^{\text{golden cross}}$	0.02	16.26*	0.49	Ten golden cross	1.218**	4.24	1.179*	-0.09	273.4**	0.08
Long low golden cross	0.03**	11.54*	0.41				• χ_{NW}^2 p -value=0.07			
Short high golden cross	0.02	-4.72	-0.06							

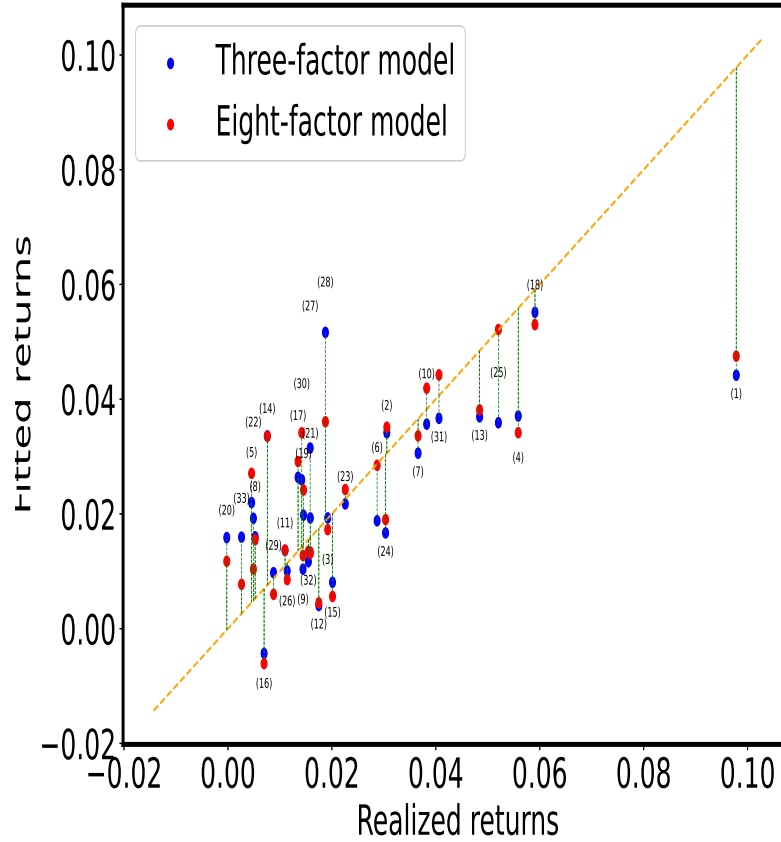


Figure 1. **Realized versus fitted returns across the 33 crypto portfolios**

Plotted are the realized returns (x-axis) and the fitted returns (y-axis) corresponding to the crypto portfolios indexed from 1 to 33. The fitted average returns are based on equation (3).

- Three-factor model comprises market, size, and momentum;
- Eight-factor model comprises market, size, reversal, lottery, skewness, downside, upside, and golden cross.

See the model estimates in Table 8. The crypto test portfolios and the corresponding model errors are shown in Table 10.